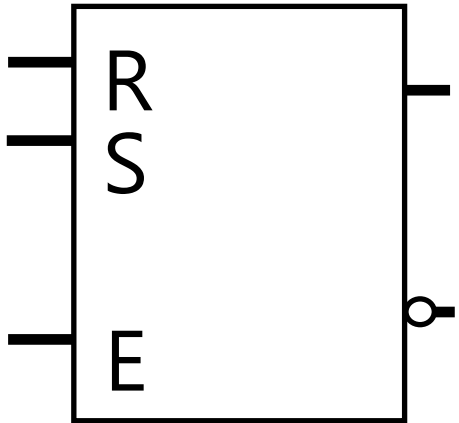
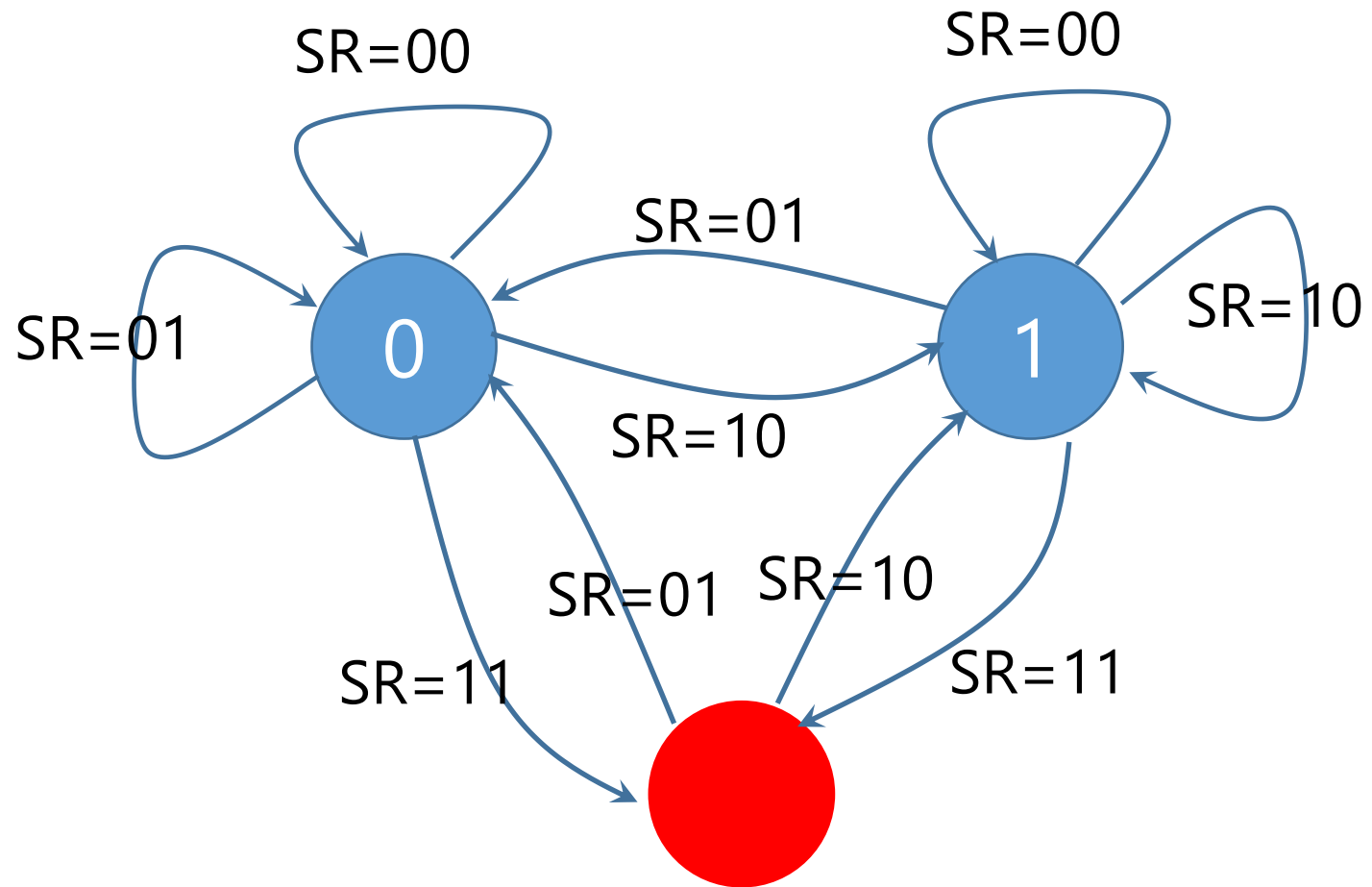

No More Lec/Lab >> But we have Lab Sections!
Bonus Feedback as Lab12

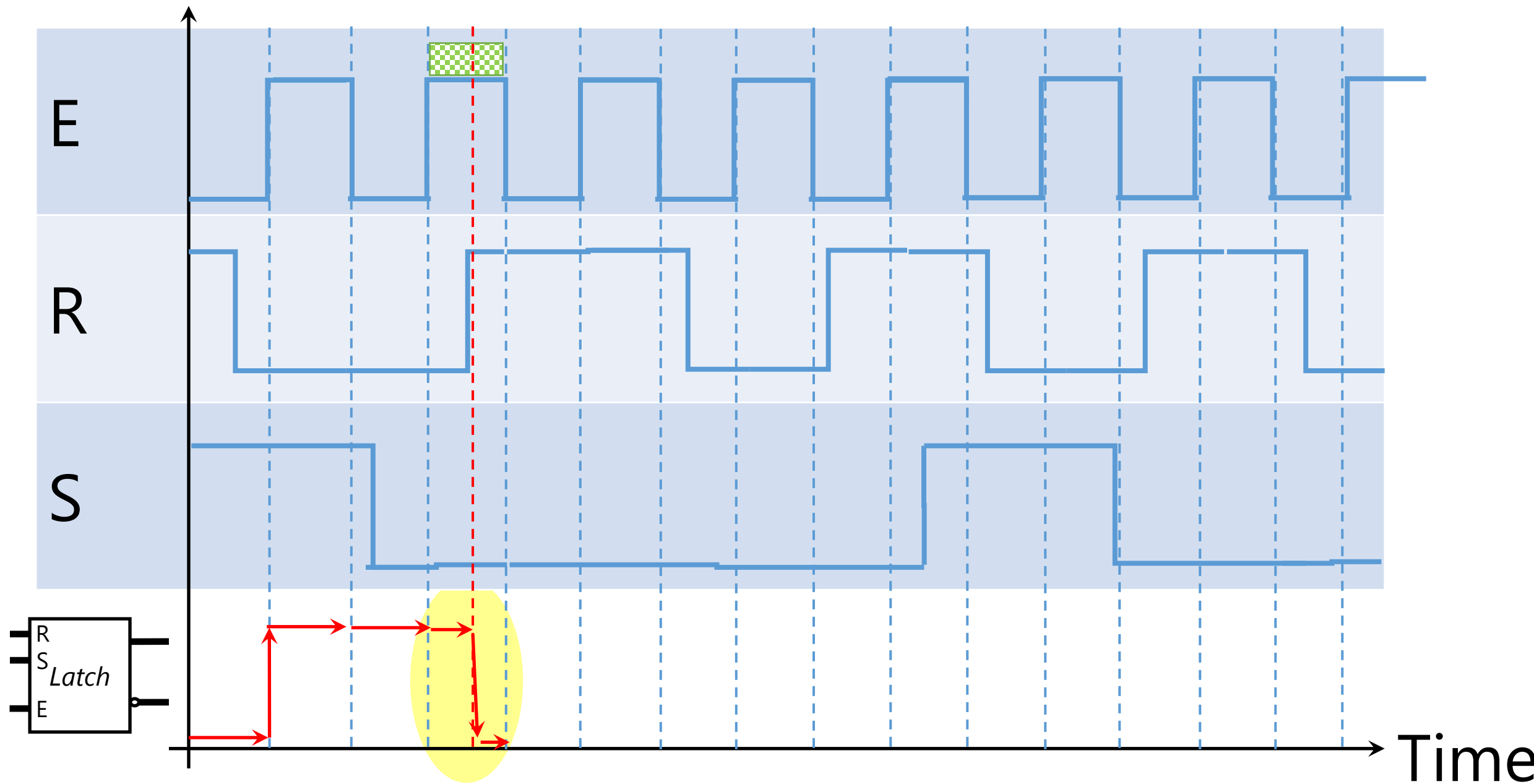
Flip-Flop



S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$



Voltage

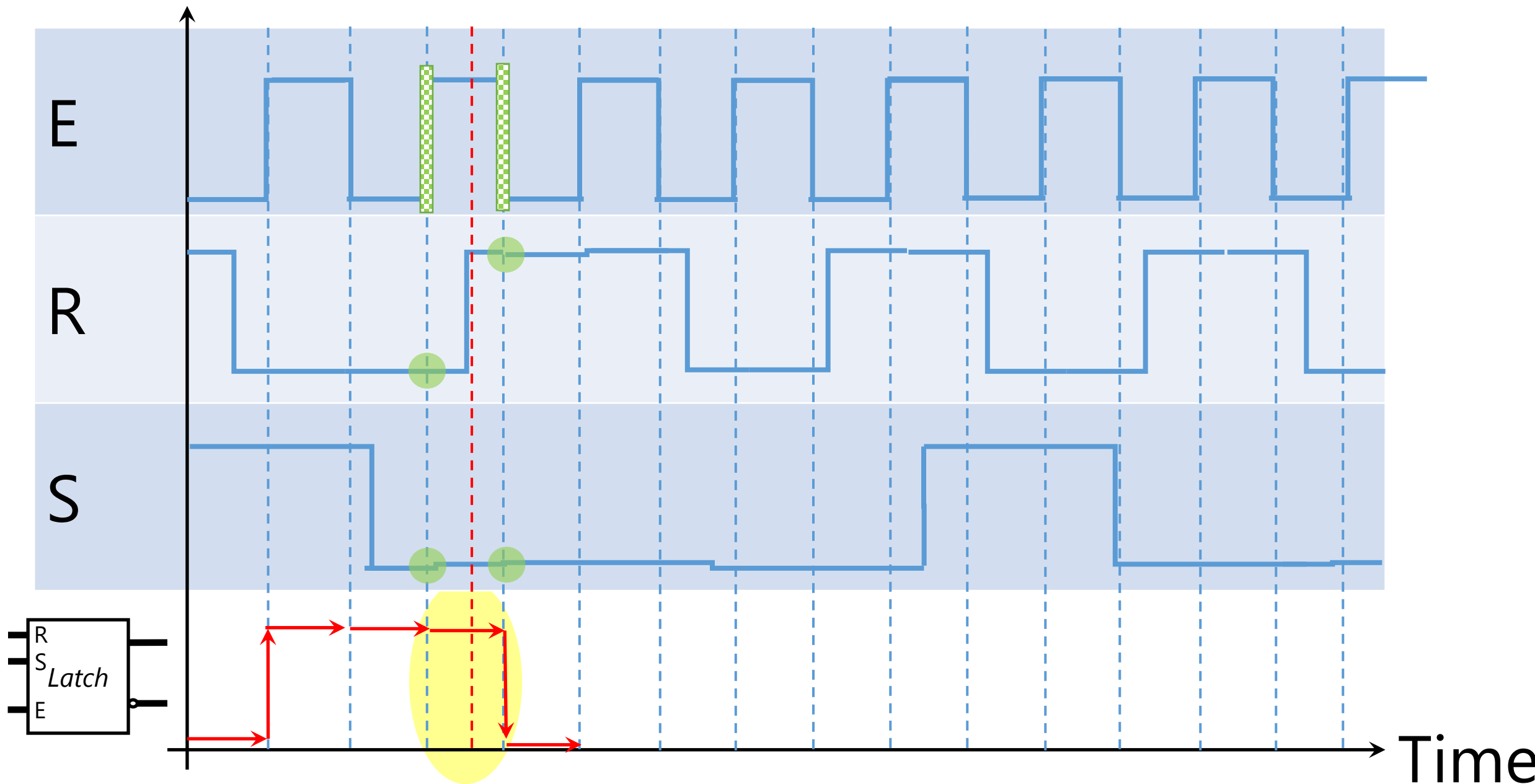


Active High (low) is controlling the change to only a specific **time period**, but is it possible to reduce it to a **moment**?



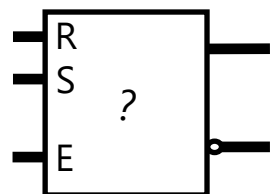
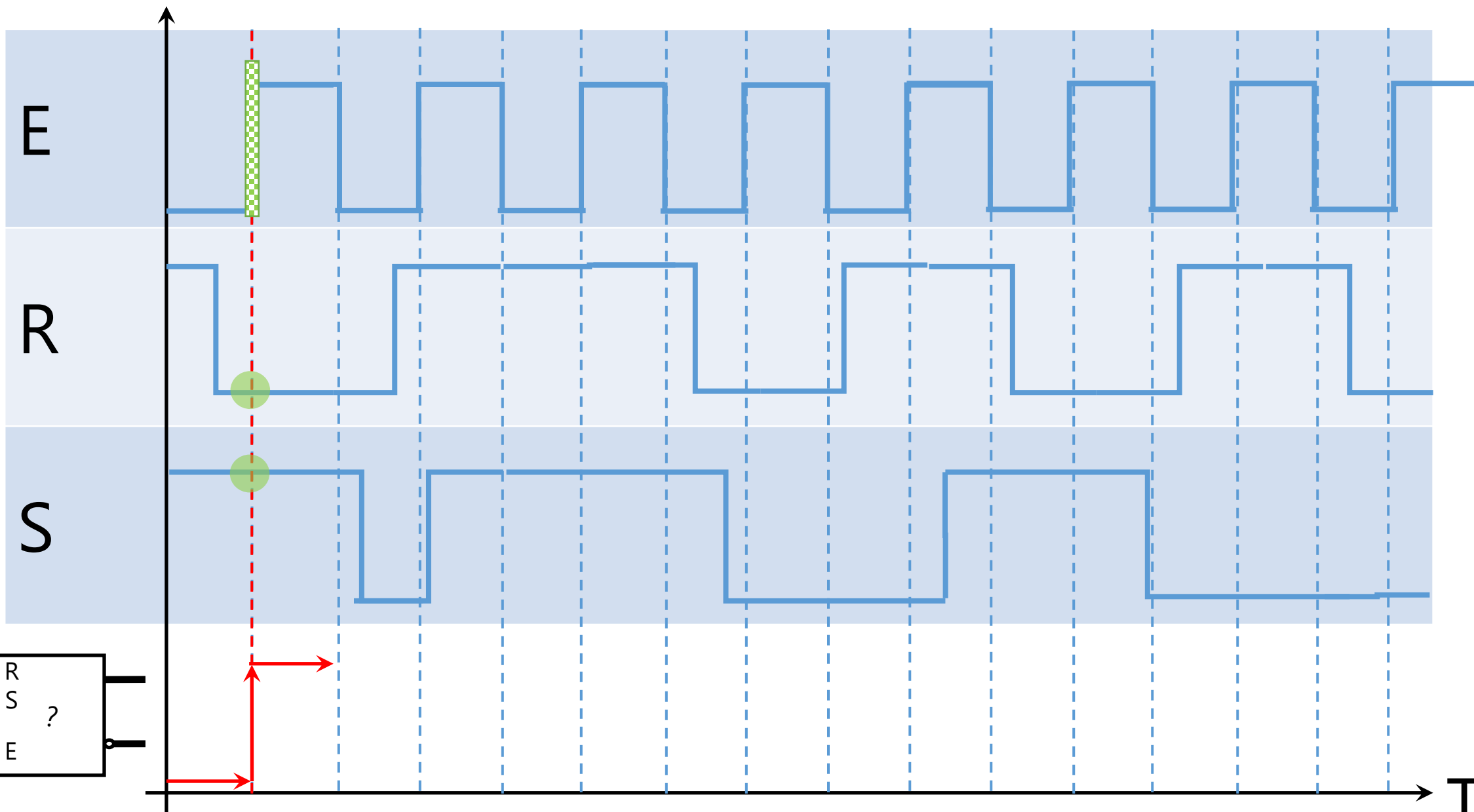
Peephole

Voltage



Level vs. Edge

Voltage



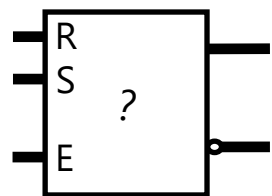
Time

Voltage

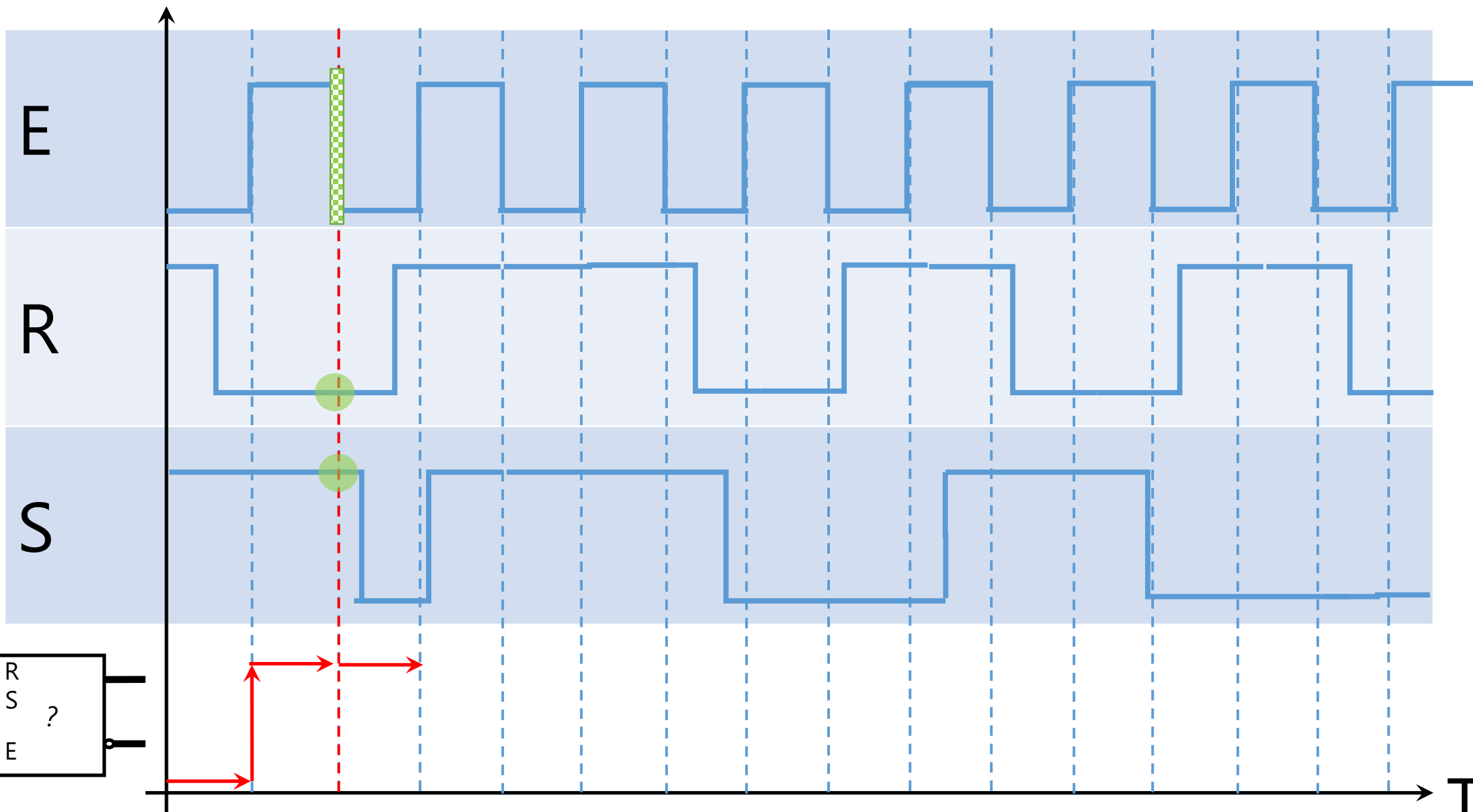
E

R

S



Time

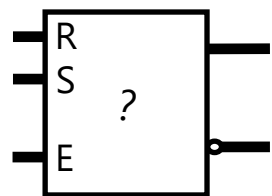


Voltage

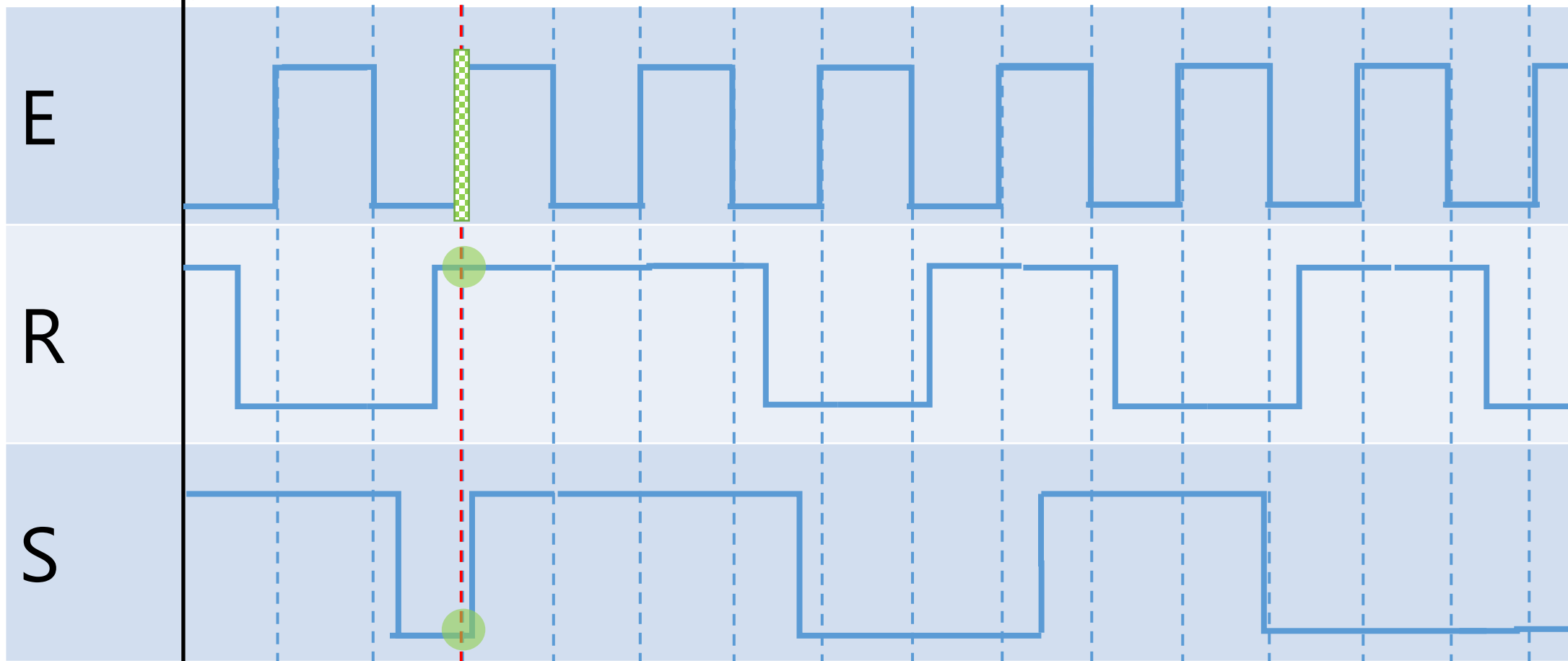
E

R

S



Time



Both Edge

From one edge to the next edge, locked!

Both vs. Single Edge

Single Edge

Positive vs. Negative



(b) Positive-edge response



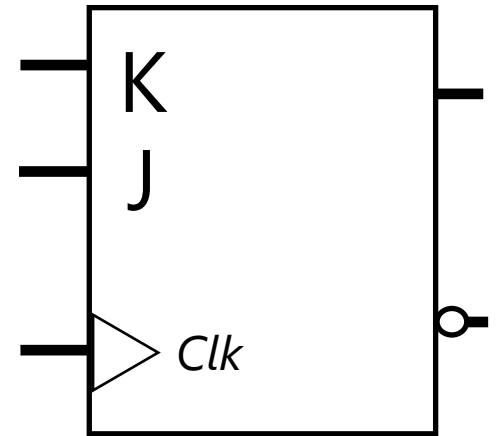
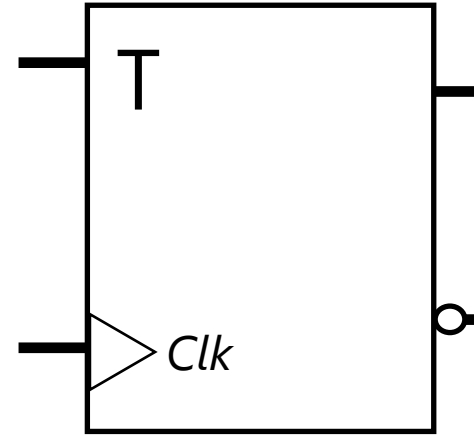
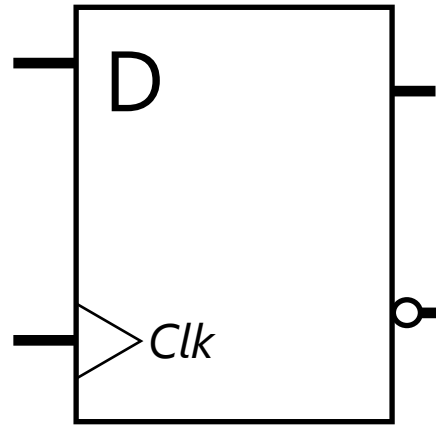
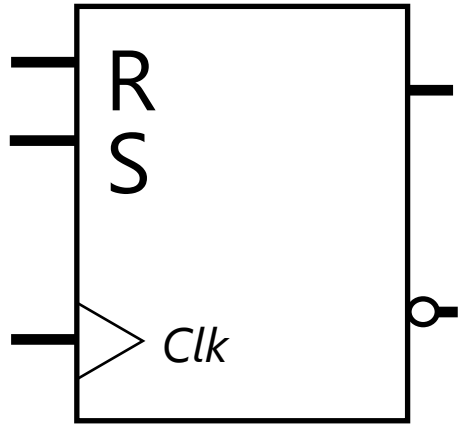
(c) Negative-edge response

FIGURE 5.8

Clock response in latch and flip-flop

Flip-Flop

A single edge enabled latch



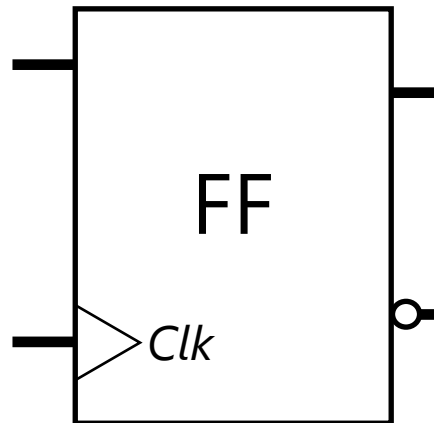
S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$

D	Q
0	0
1	1

T	Q
0	Q_t
1	Q'_t

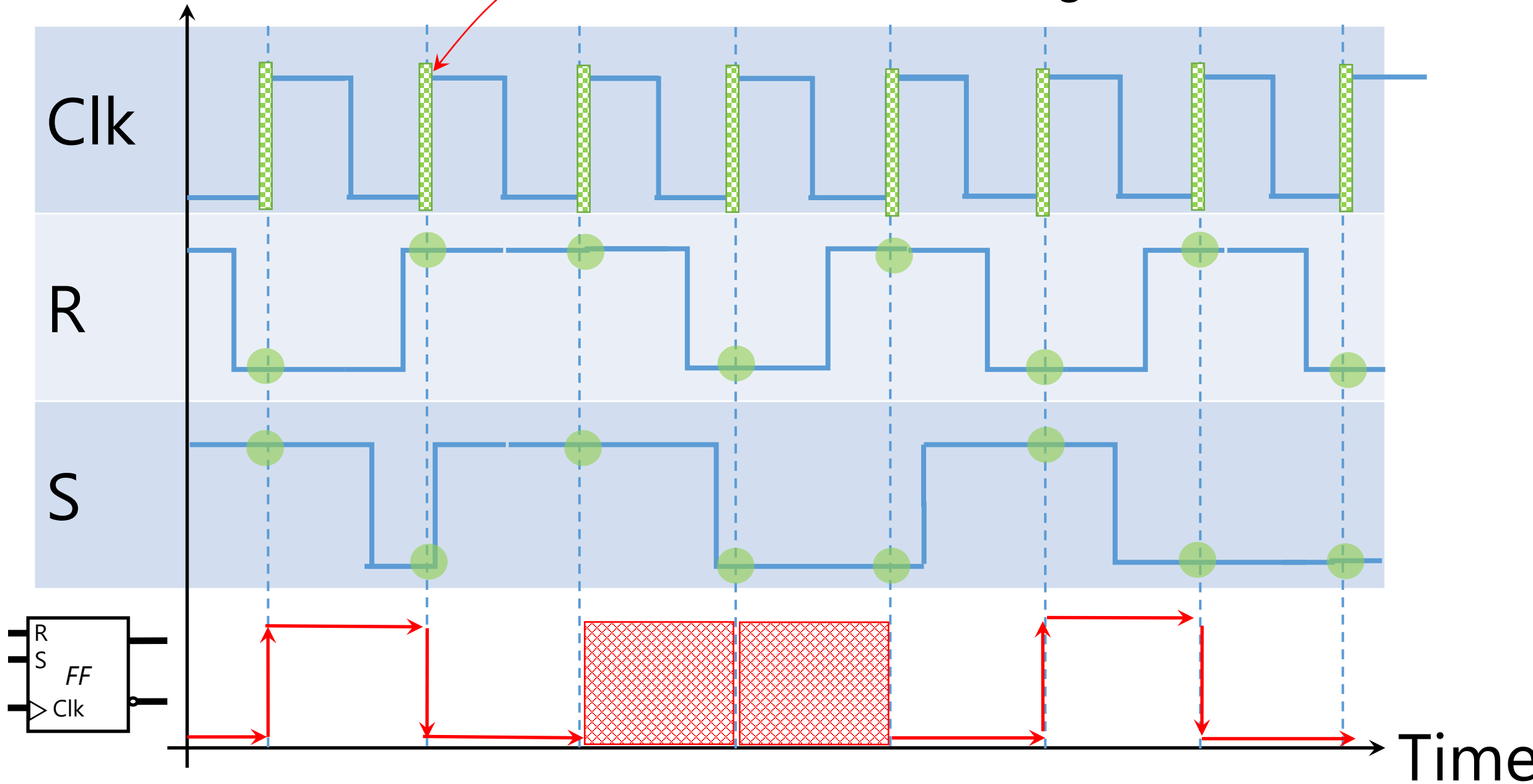
J	K	Q
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

Single Edge *Positive*

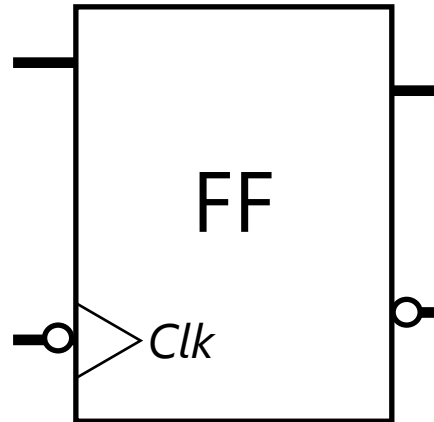


Voltage

Positive Edge $0 \rightarrow 1$



Single Edge *Negative*



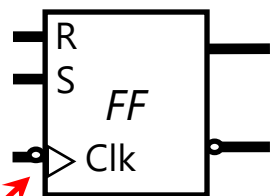
Voltage

Negative Edge $1 \rightarrow 0$

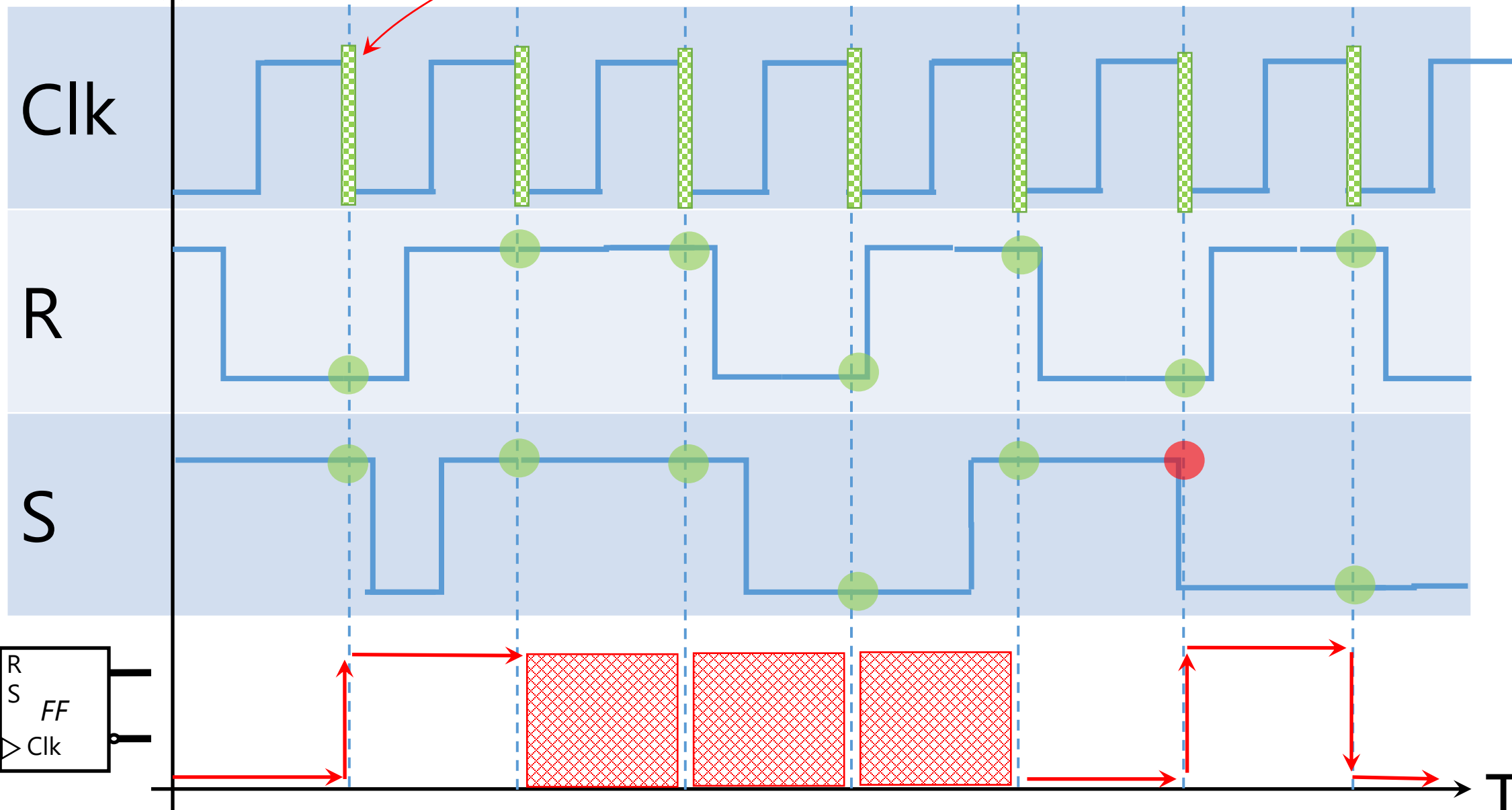
Clk

R

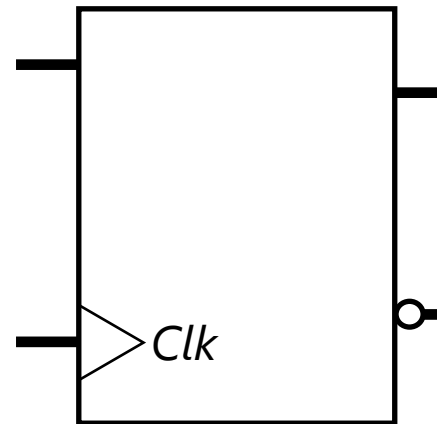
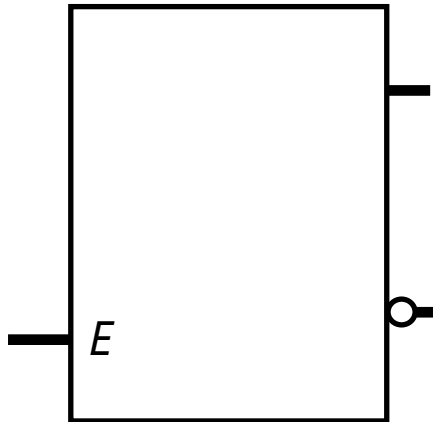
S



Time

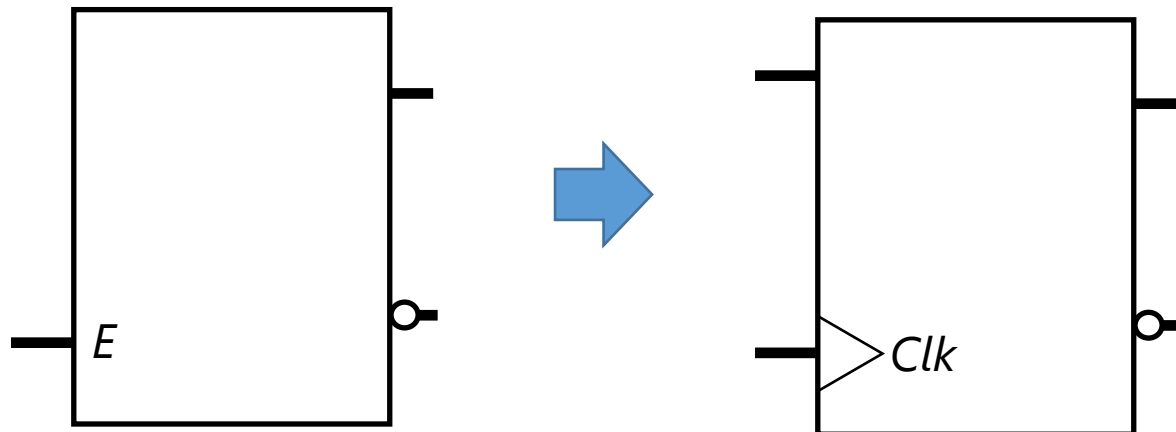


Latch → Flip-Flop



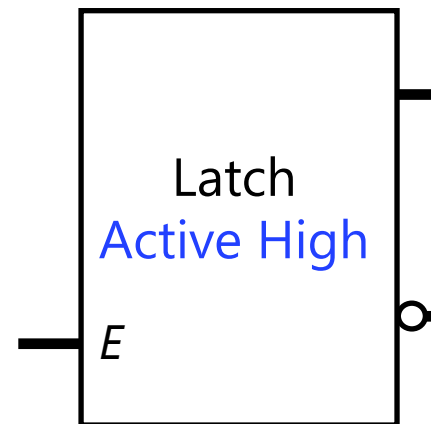
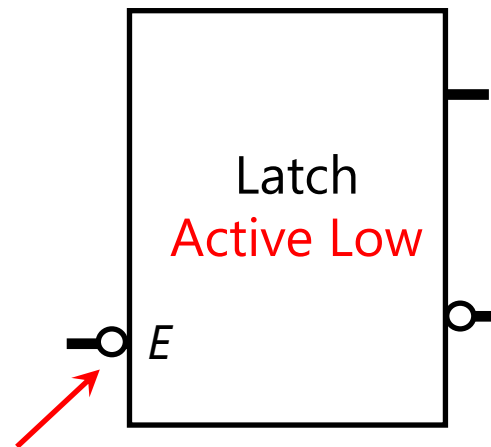
Latch → Flip-Flop

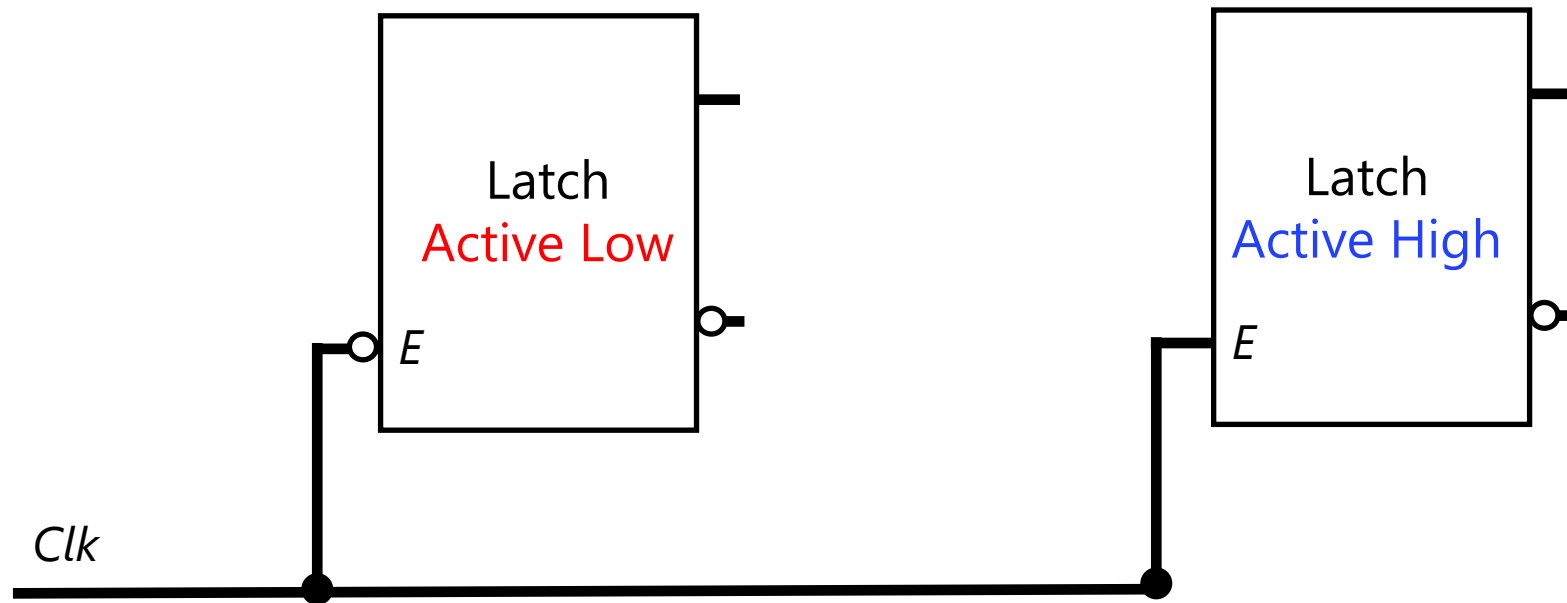
How to Design it?

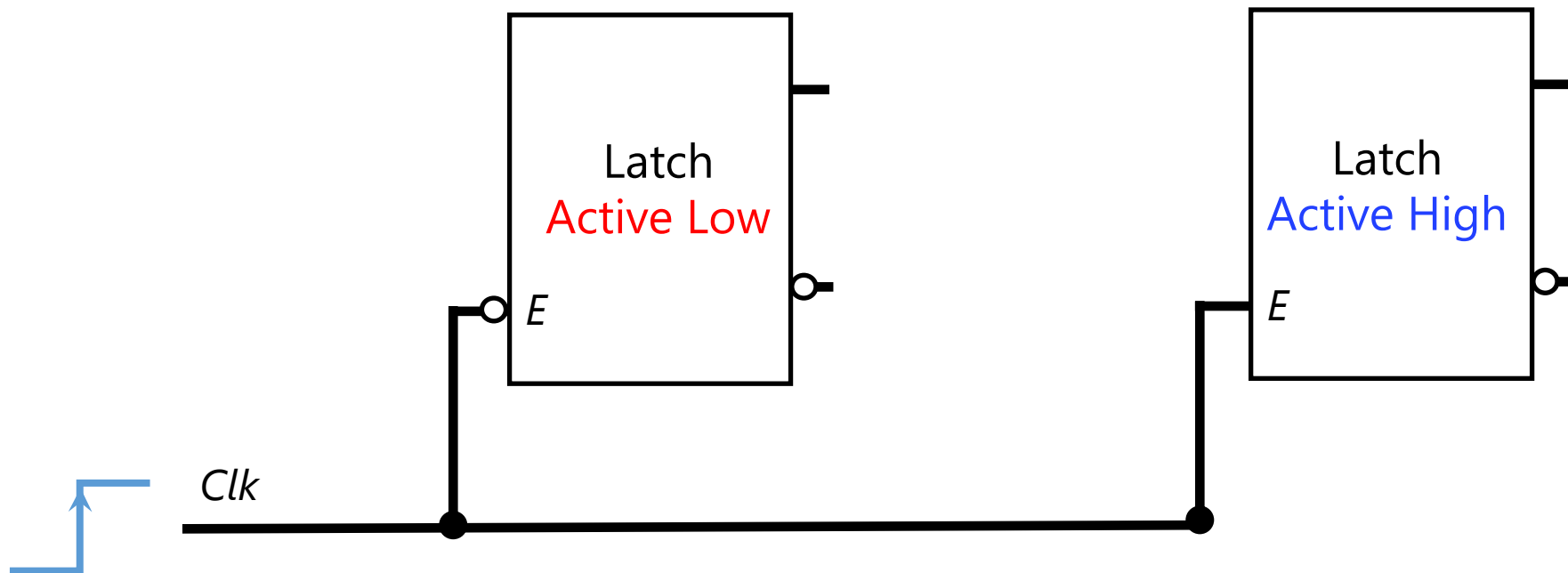


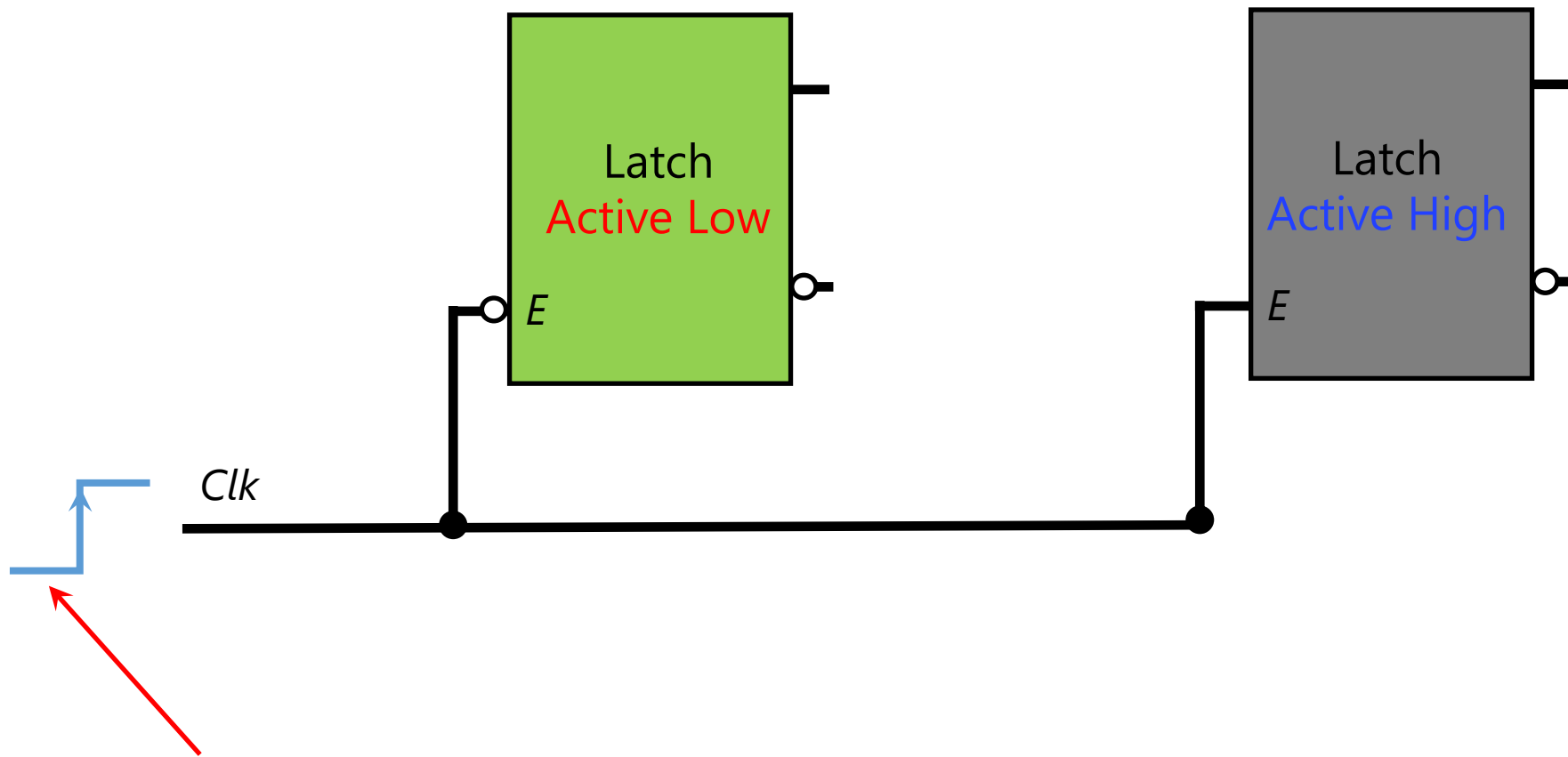


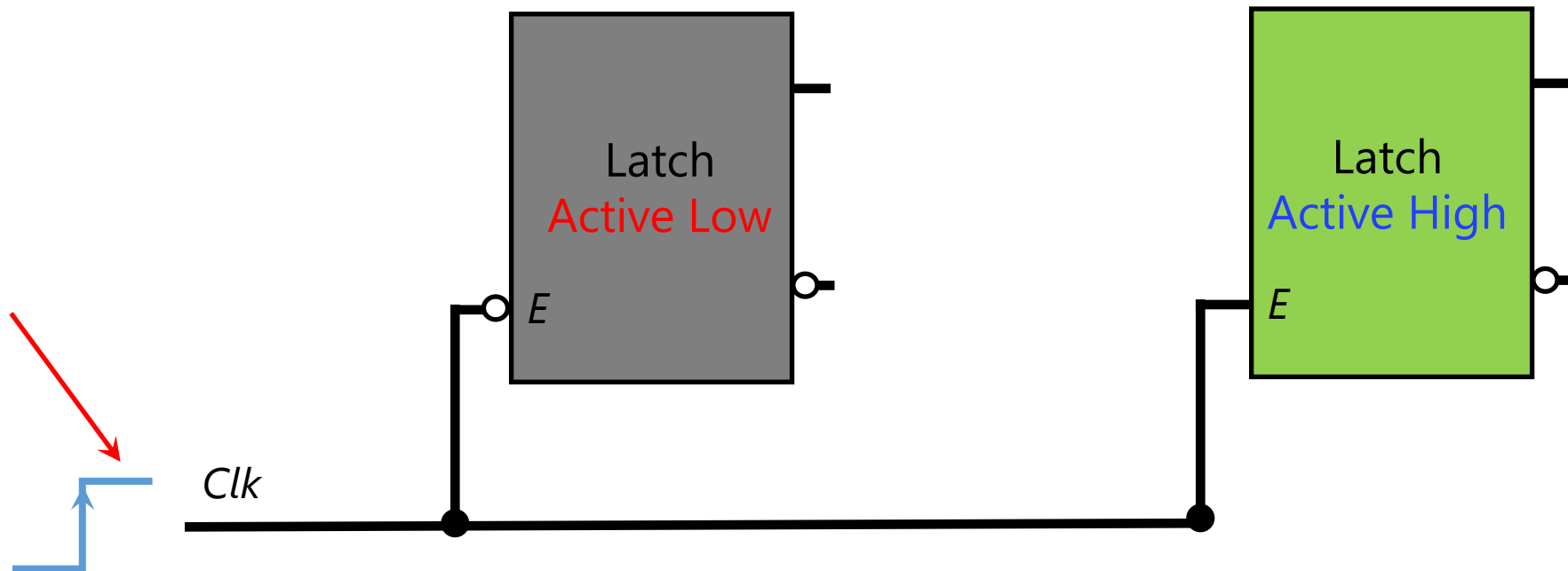
This door in my hotel bathroom can close off either of 2 doorways.
https://www.reddit.com/r/mildlyinteresting/comments/anz5z8/this_door_in_my_hotel_bathroom_can_close_off/





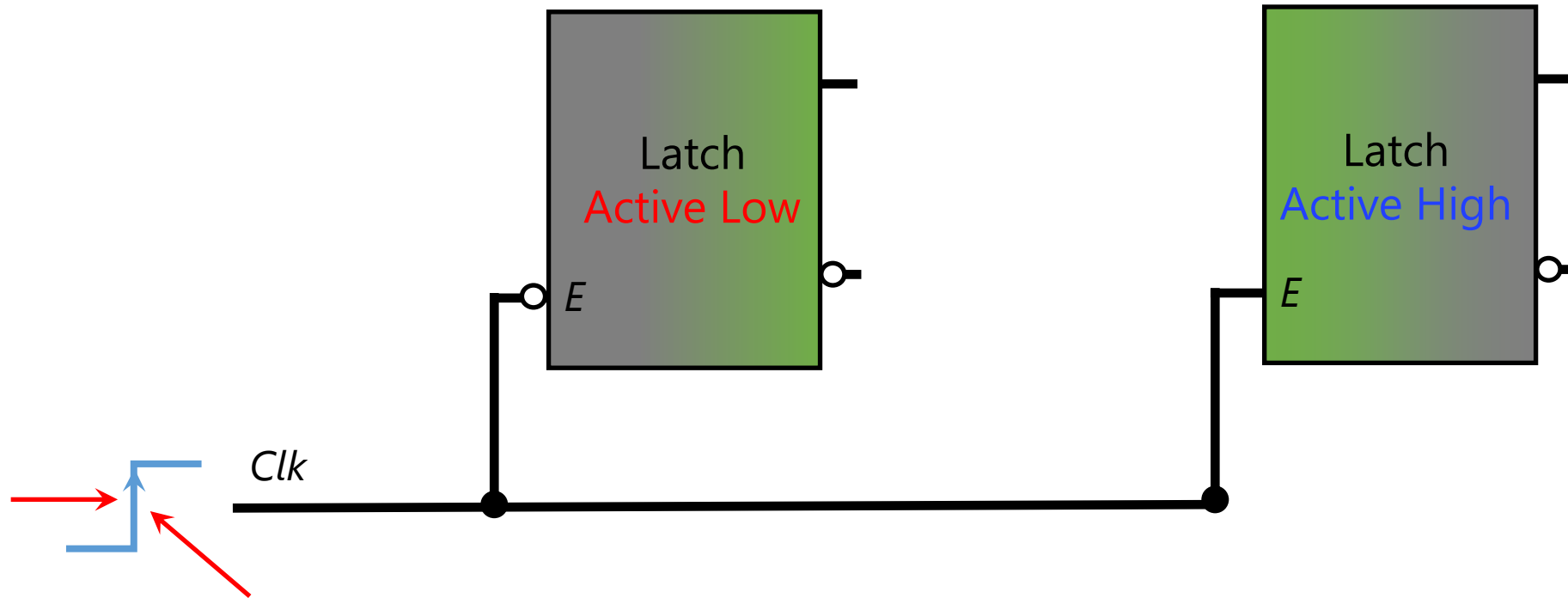






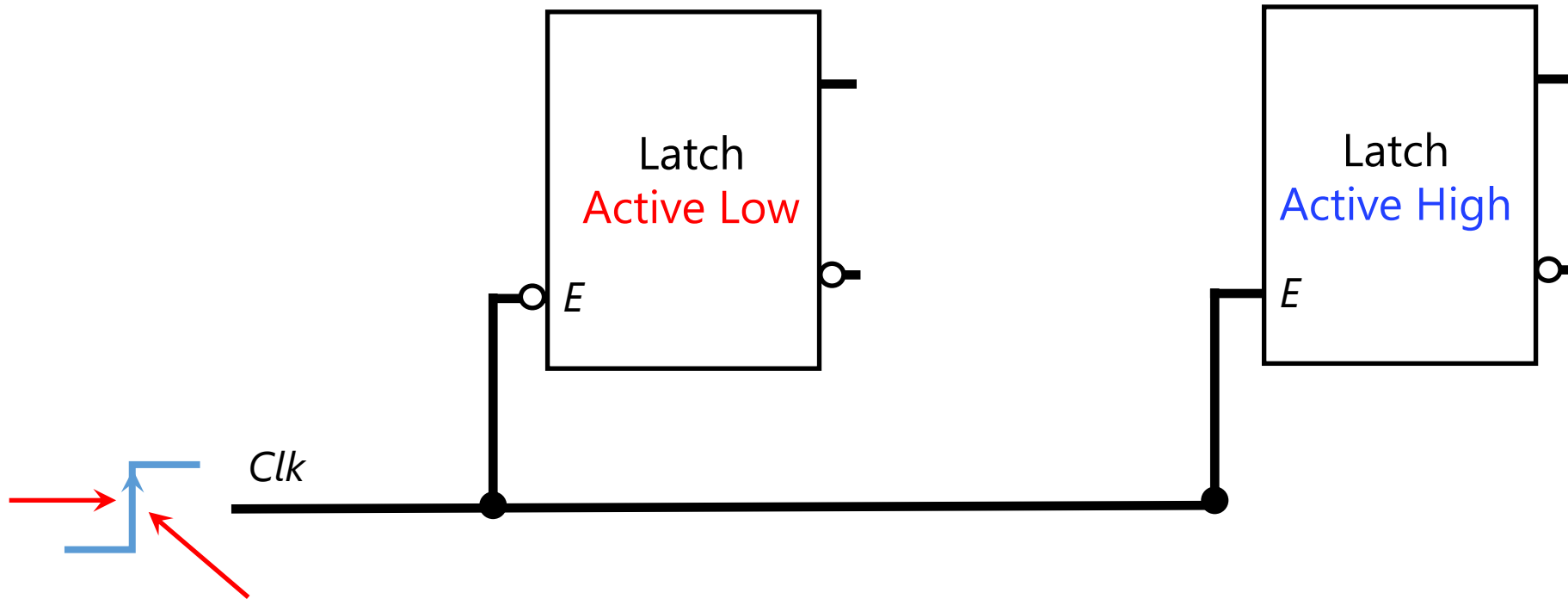
Opening the first → Closing the second
Closing the first → Opening the second





One "quick" moment that the first latch is becoming stable and closing the door!
Instead, the second latch is going to accept change.





One "quick" moment that the first latch is becoming stable and closing the door!
Instead, the second latch is going to accept change.

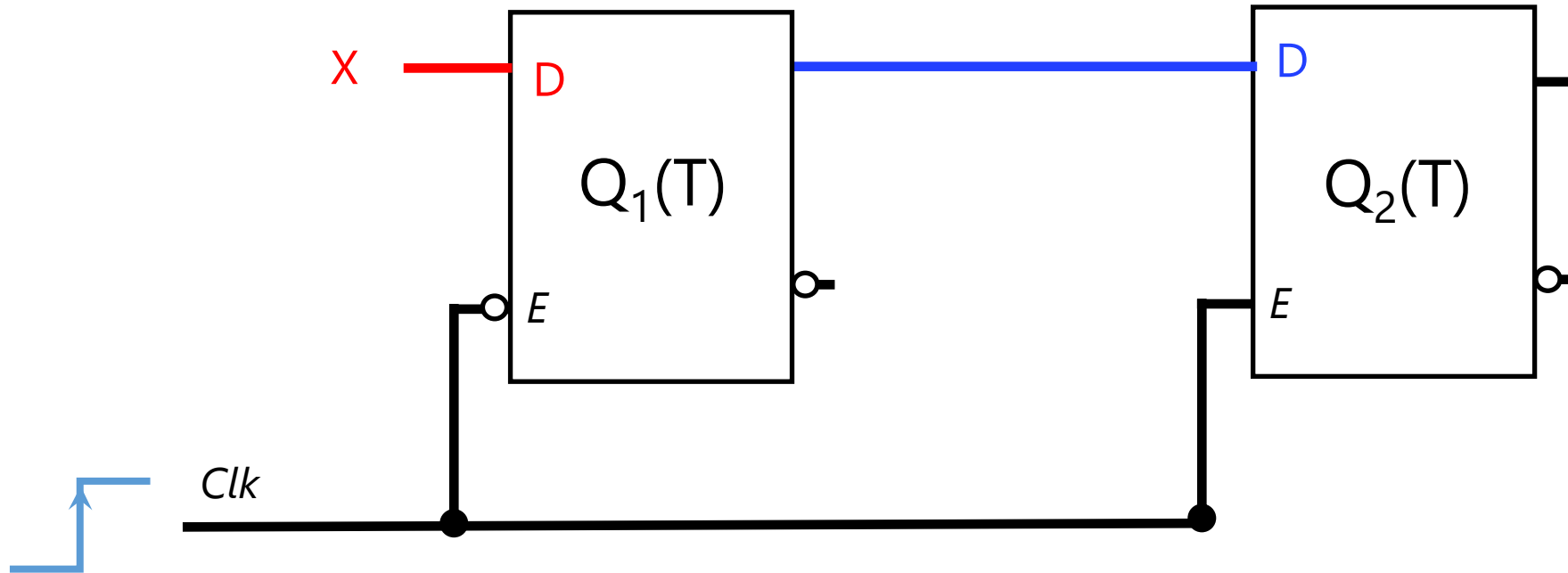
Let's force the second latch accept change from the first latch only!



Remember: In D latch, when enabled, whatever in input changes the state:

$X=0 \rightarrow Q=0$

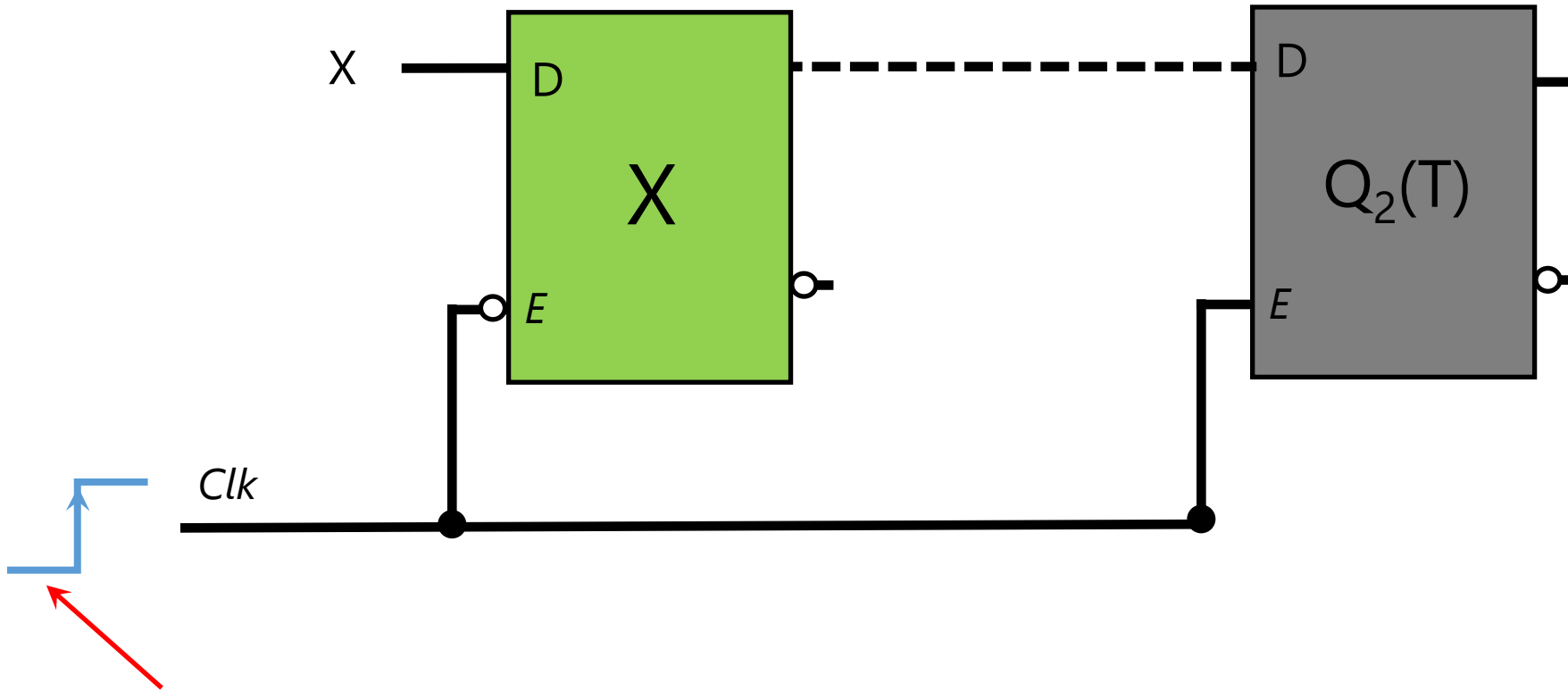
$X=1 \rightarrow Q=1$



Remember: In D latch, when enabled, whatever in input changes the state:

$$X=0 \rightarrow Q=0$$

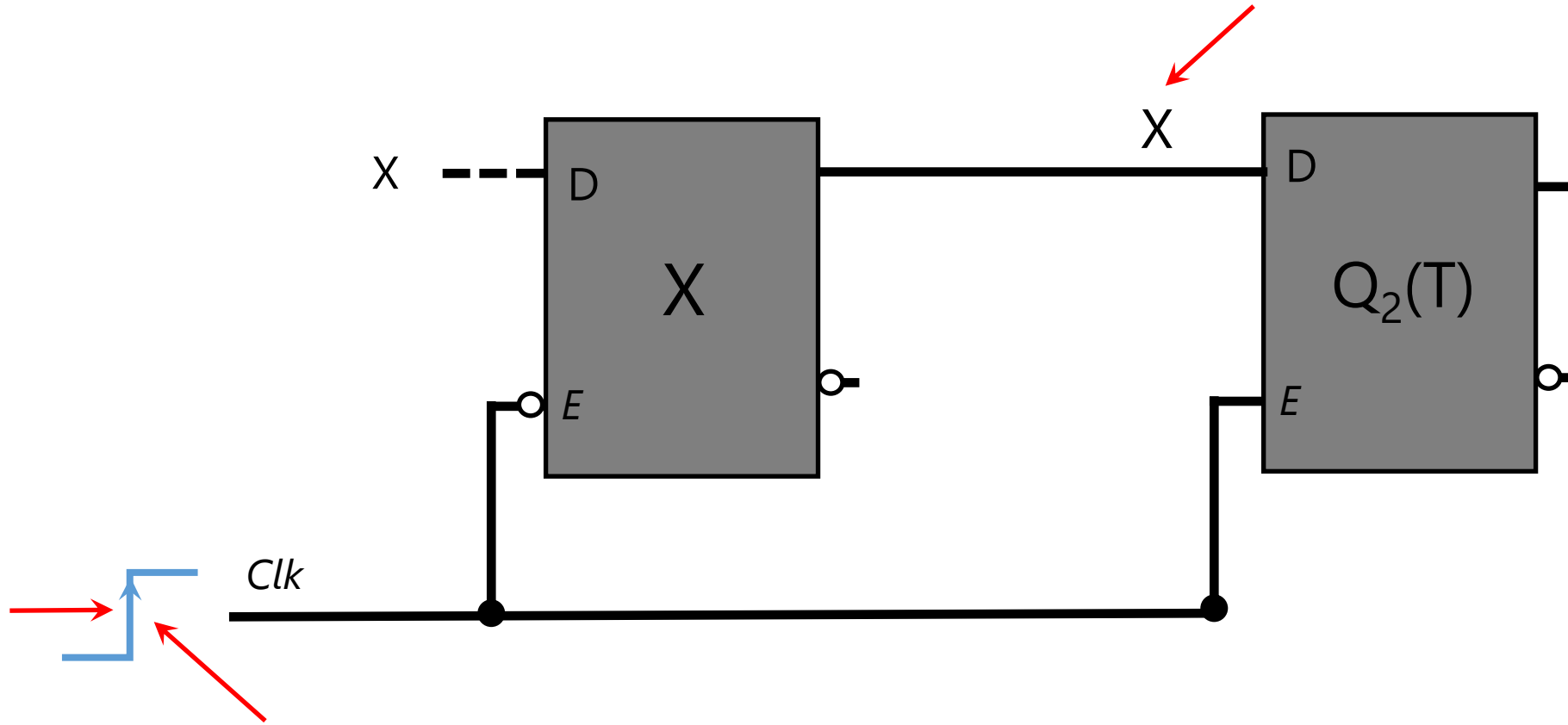
$$X=1 \rightarrow Q=1$$



Remember: In D latch, when enabled, whatever in input changes the state:

$$X=0 \rightarrow Q=0$$

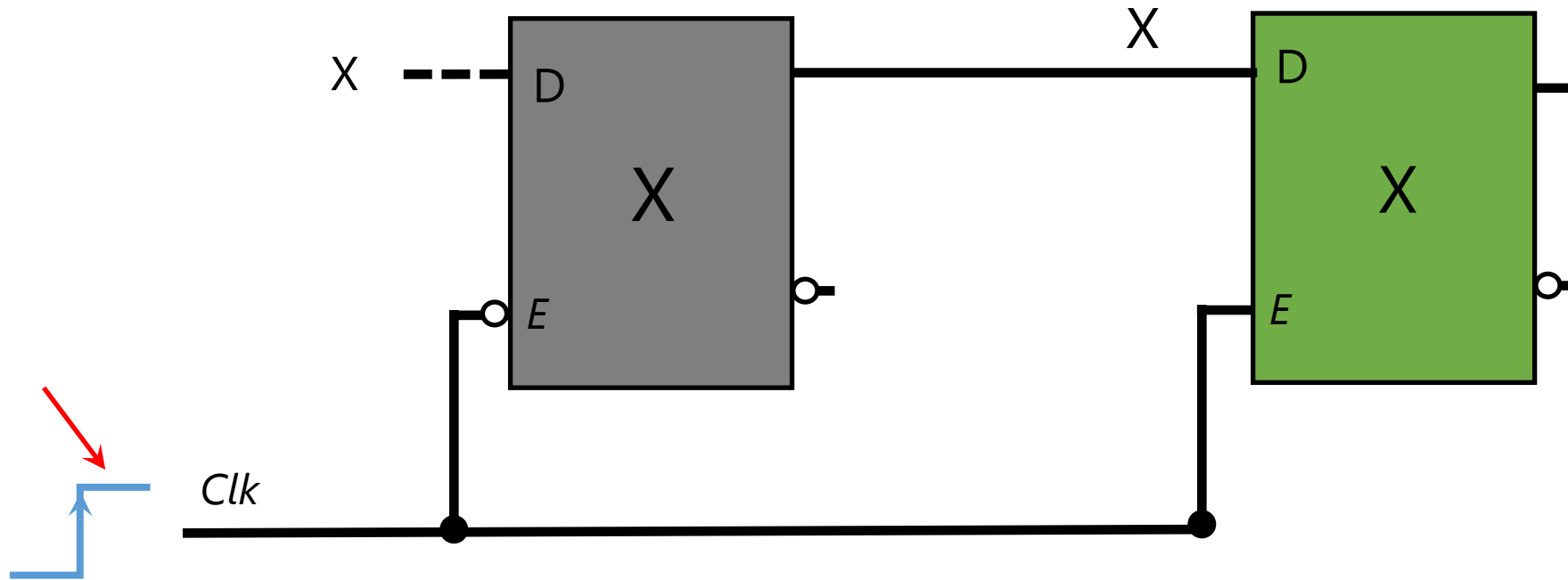
$$X=1 \rightarrow Q=1$$



Remember: In D latch, when enabled, whatever in input changes the state:

$X=0 \rightarrow Q=0$

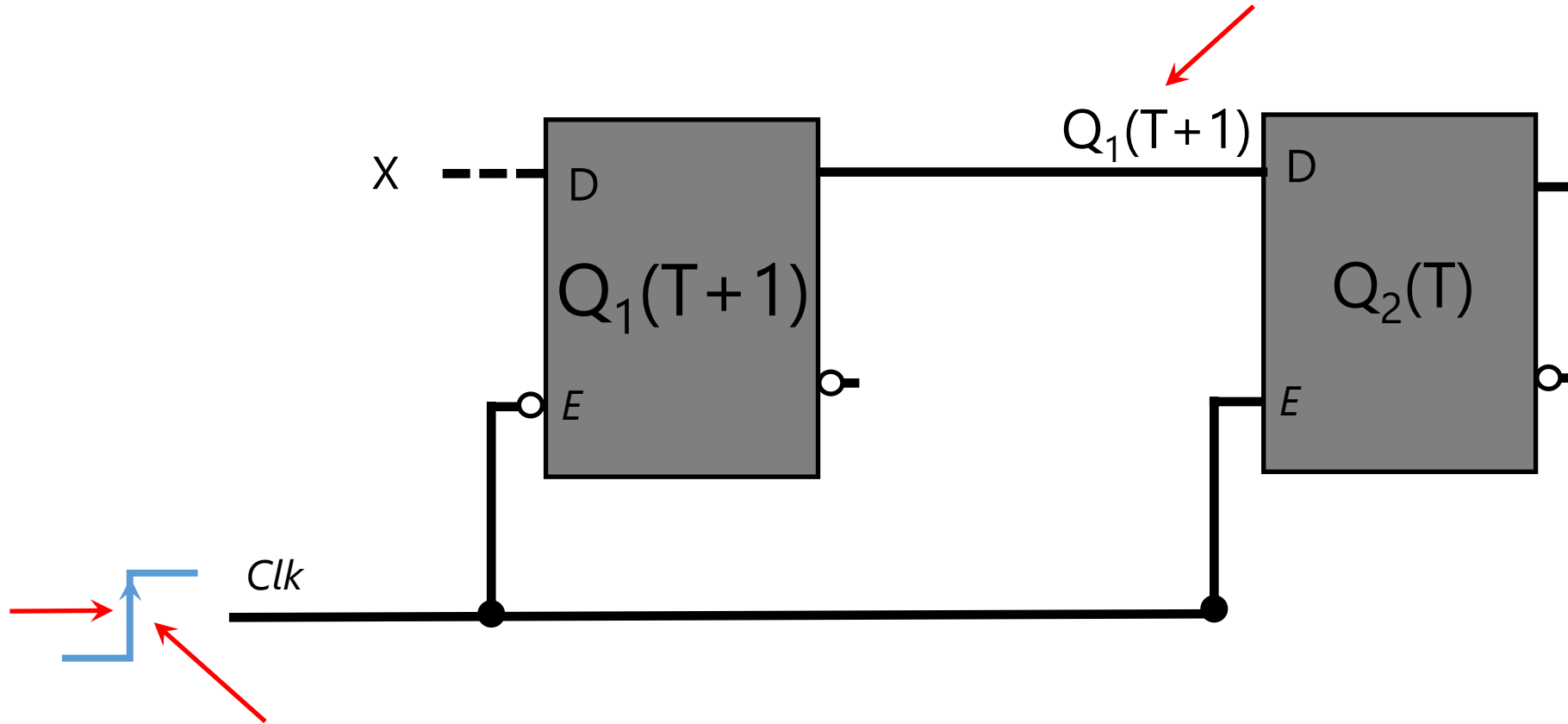
$X=1 \rightarrow Q=1$



Remember: In D latch, when enabled, whatever in input changes the state:

$X=0 \rightarrow Q=0$

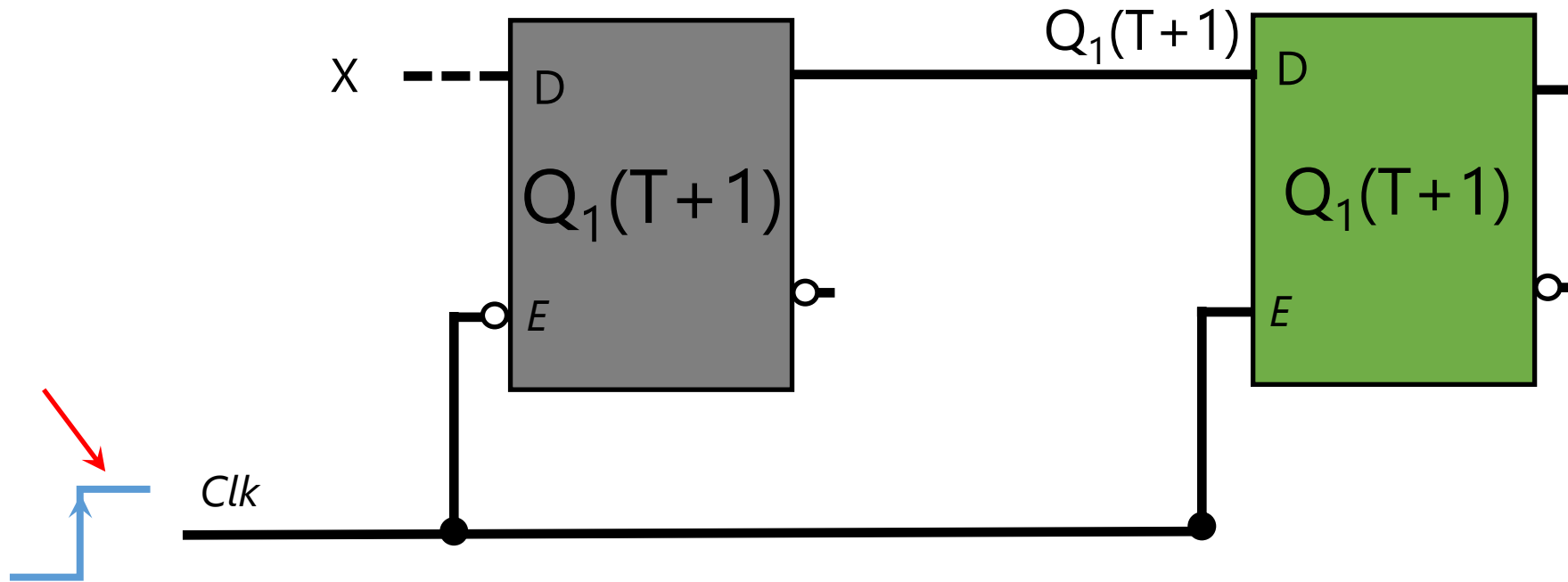
$X=1 \rightarrow Q=1$



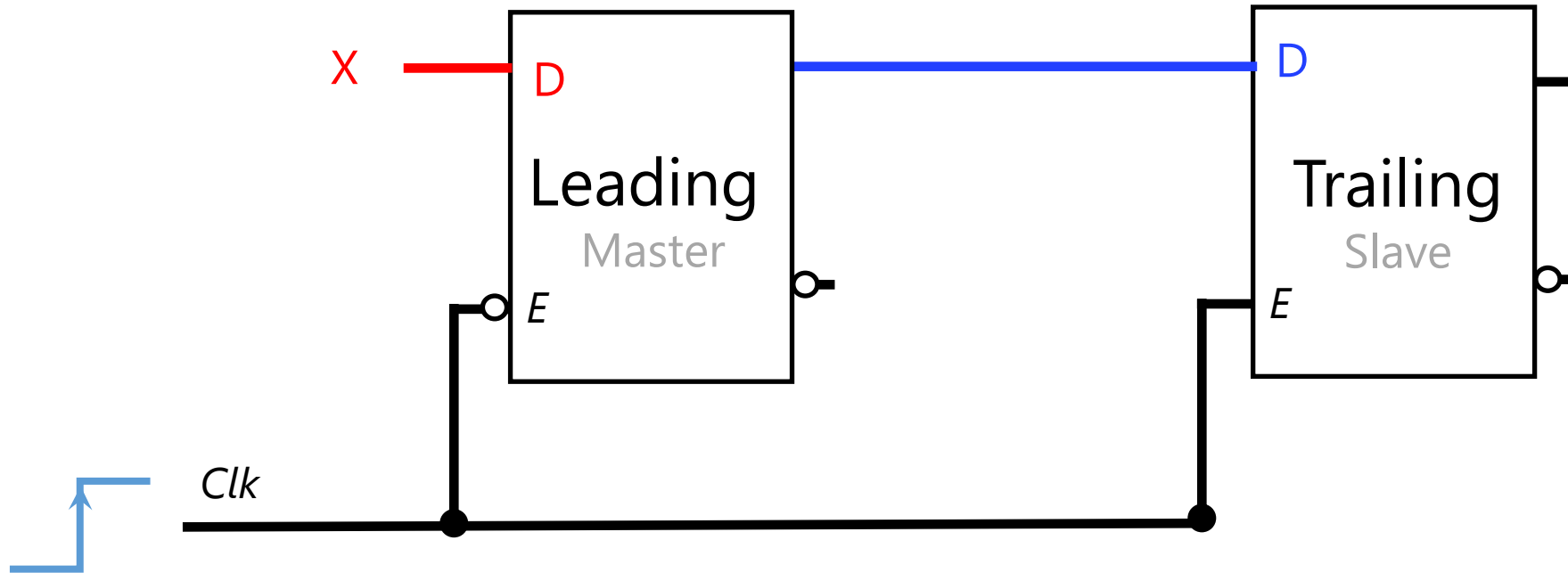
Remember: In D latch, when enabled, whatever in input changes the state:

$$X=0 \rightarrow Q=0$$

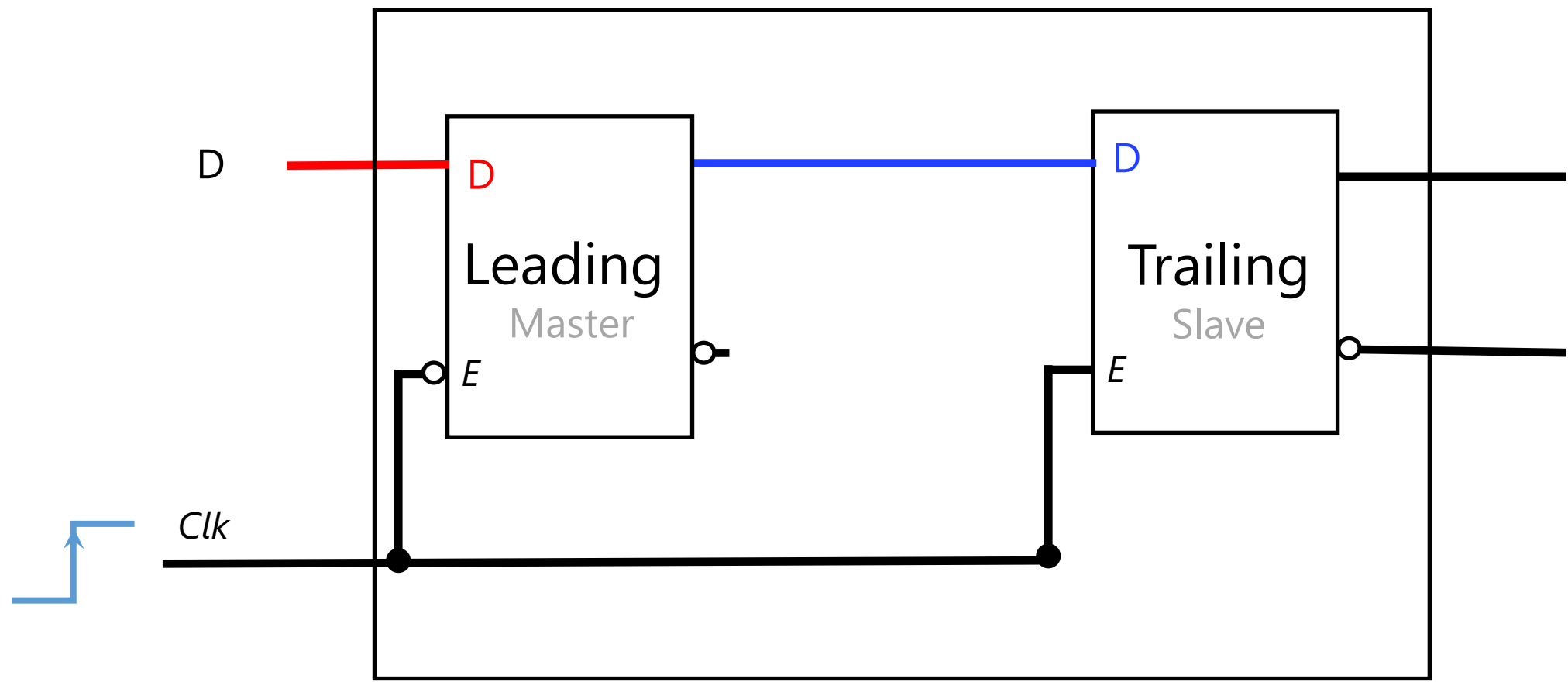
$$X=1 \rightarrow Q=1$$



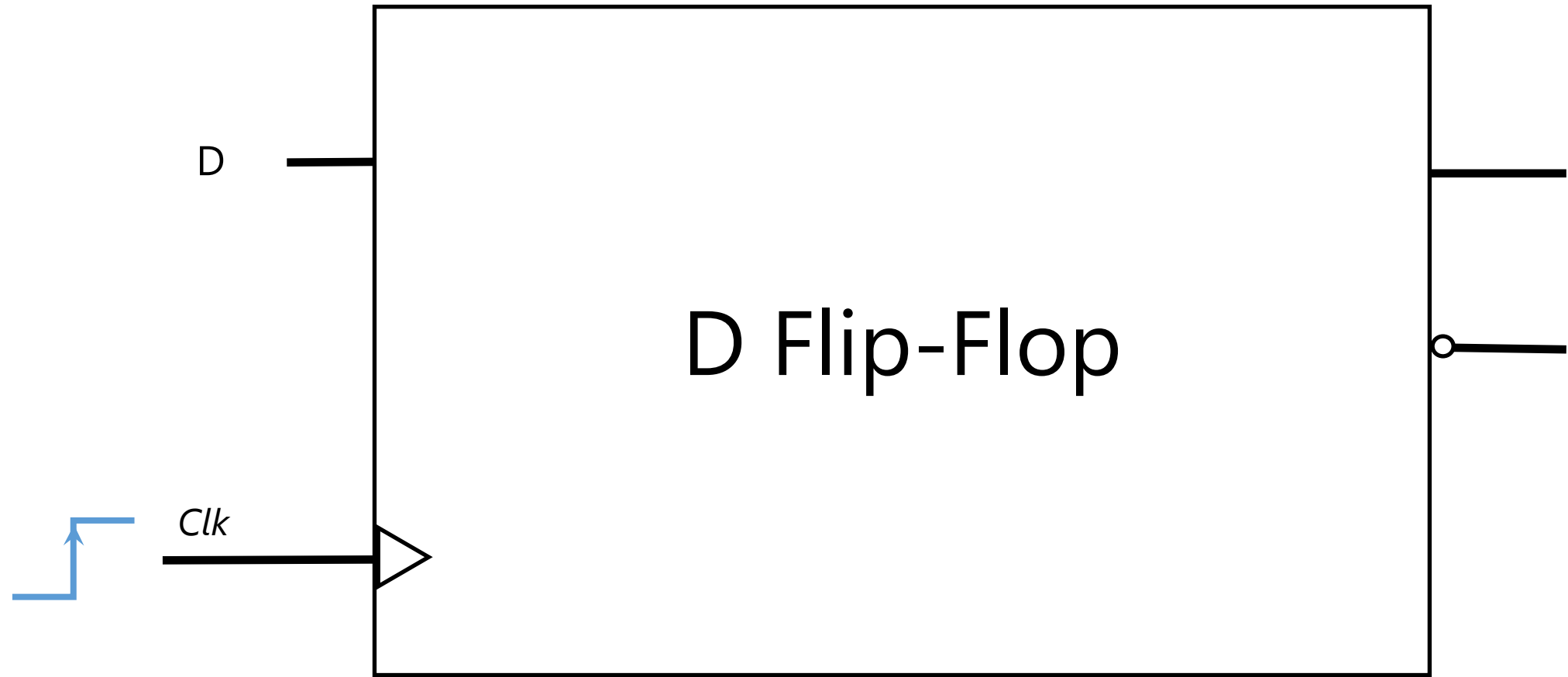
Leading-Trailing → Primary-Secondary → Two-Stage FF → Edge-Trigger
Master-Slave

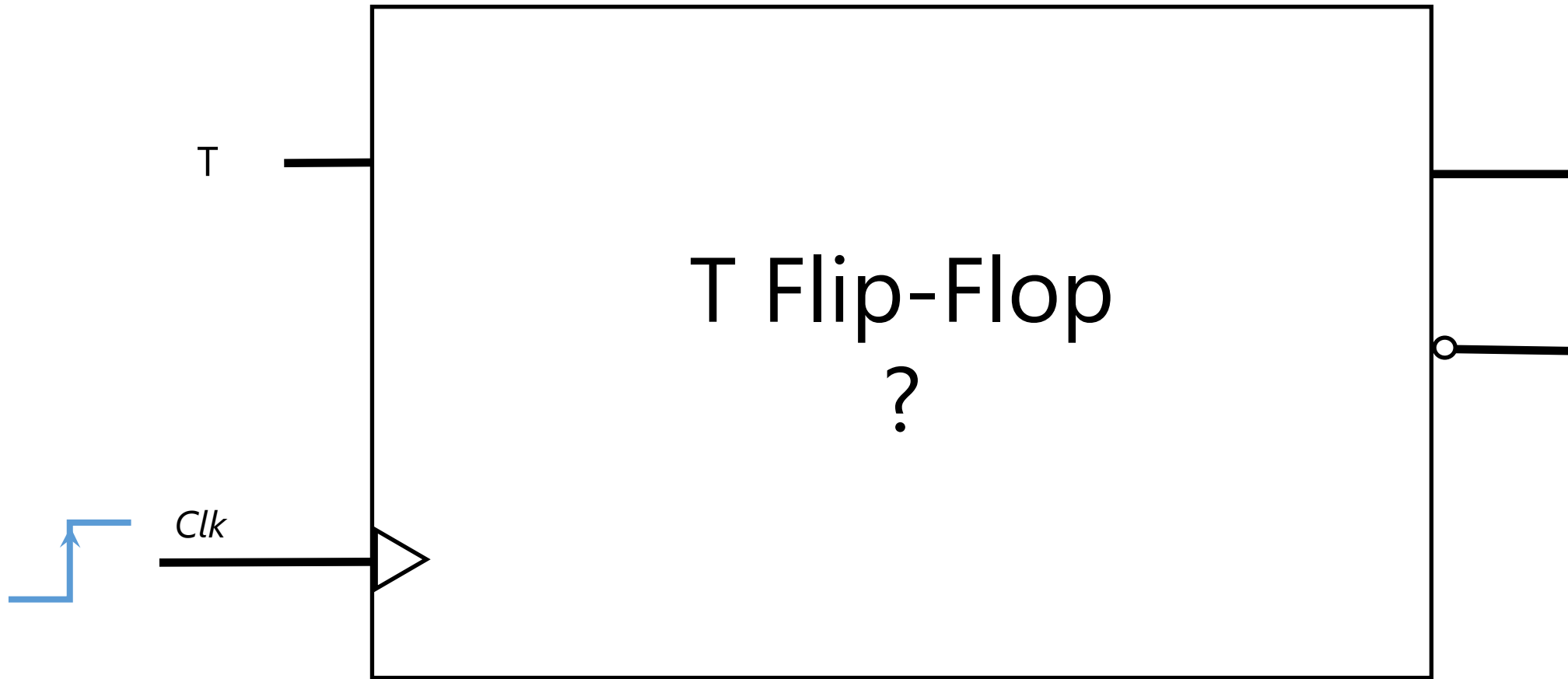


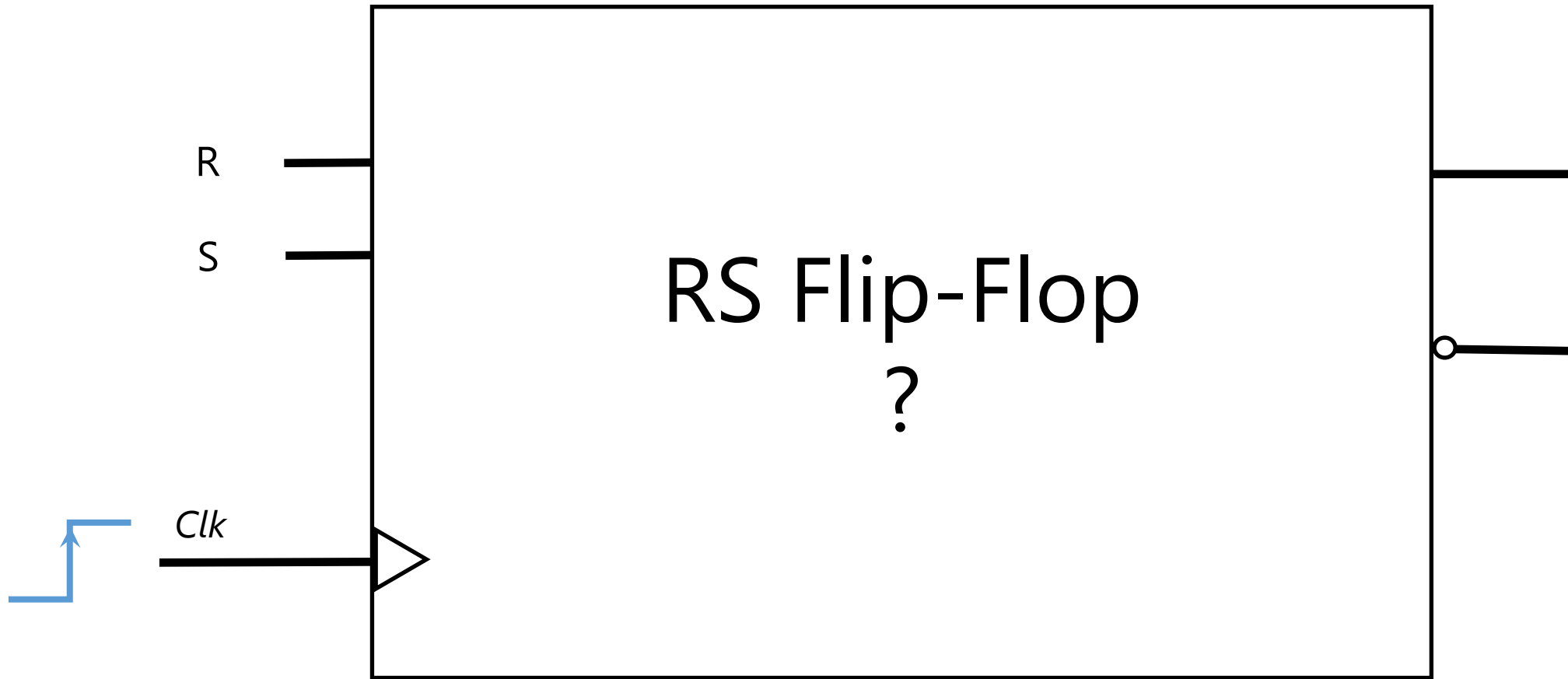
Leading-Trailing → Primary-Secondary → Two-Stage FF → Edge-Trigger
Master-Slave

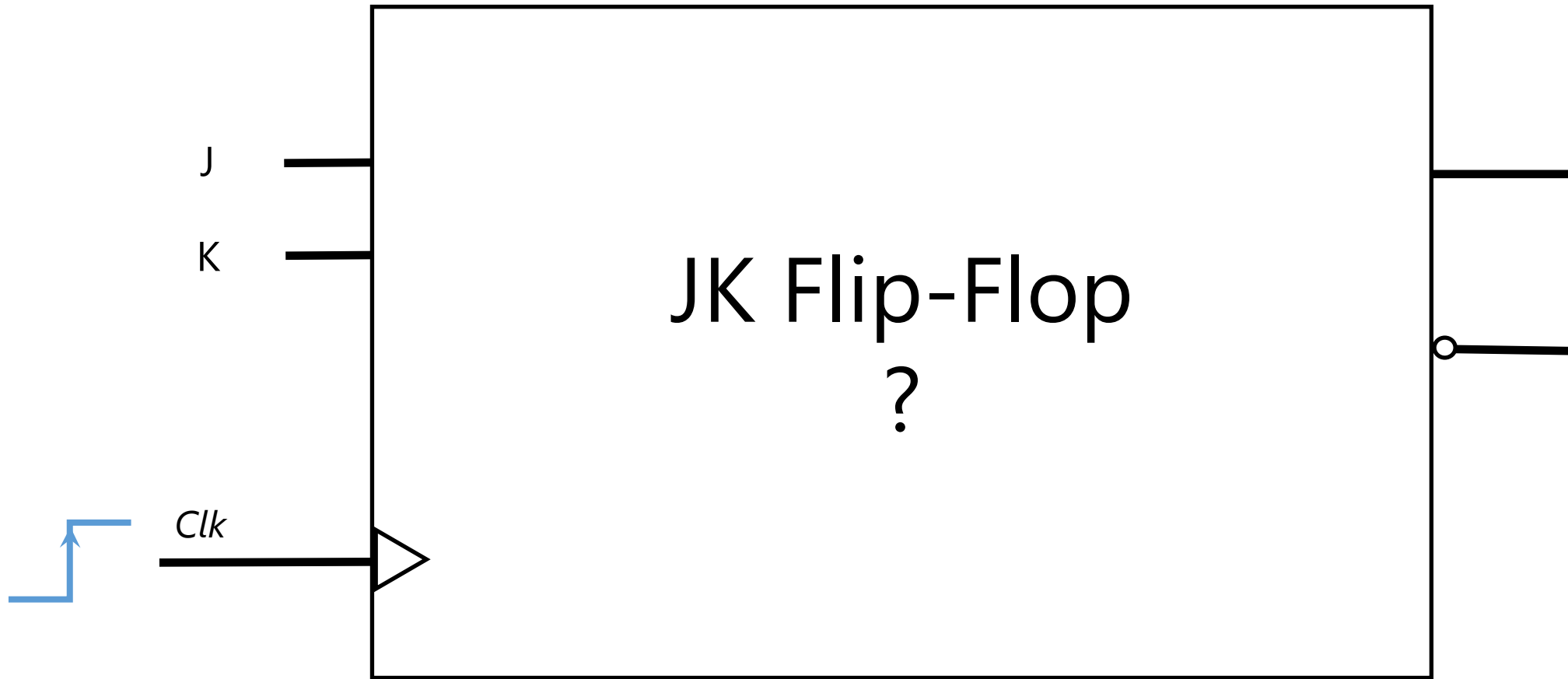


Leading-Trailing → Primary-Secondary → Two-Stage FF → Edge-Trigger
Master-Slave





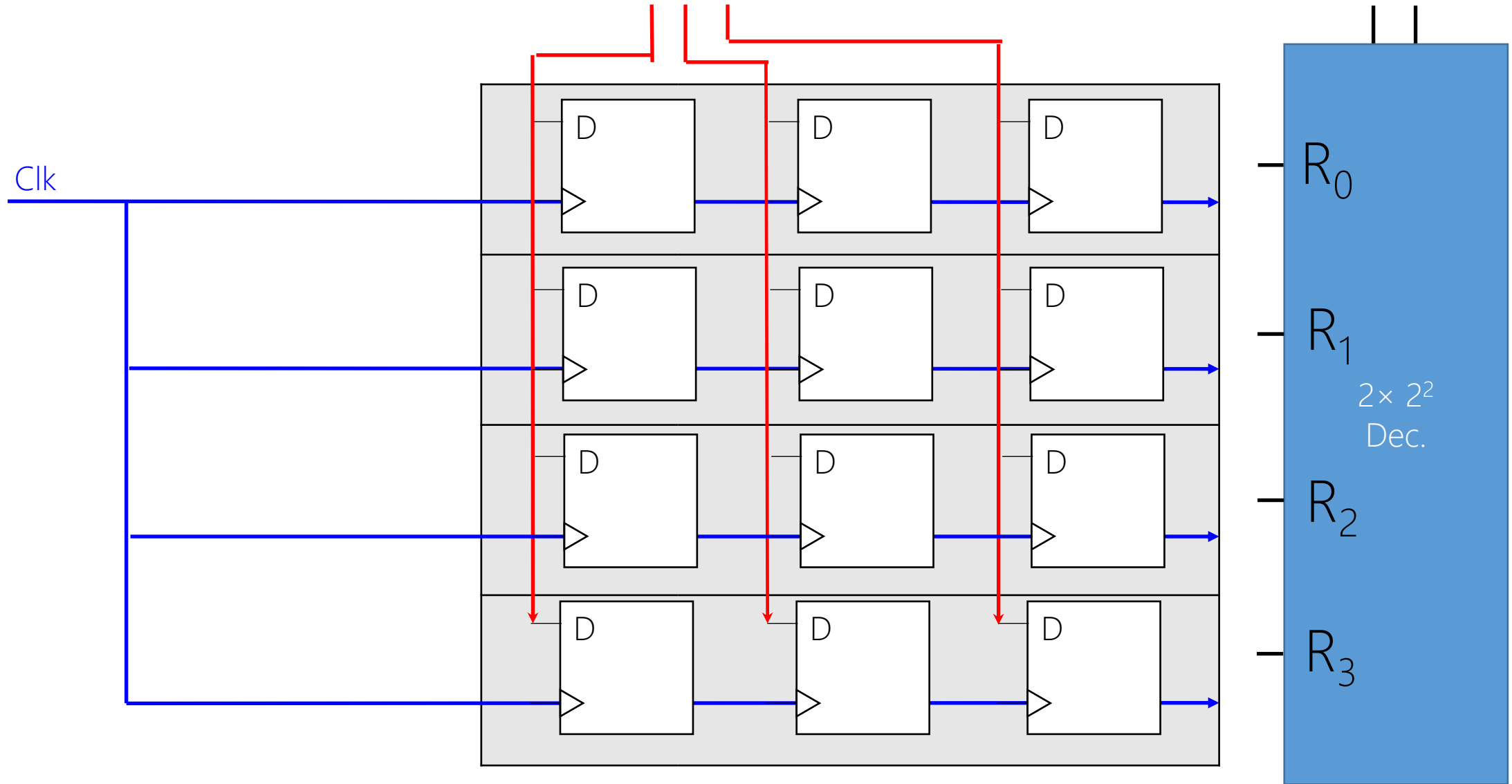




We have our ideal memory unit: Flip-Flop

3-bit Data Bus

2-bit Address Bus

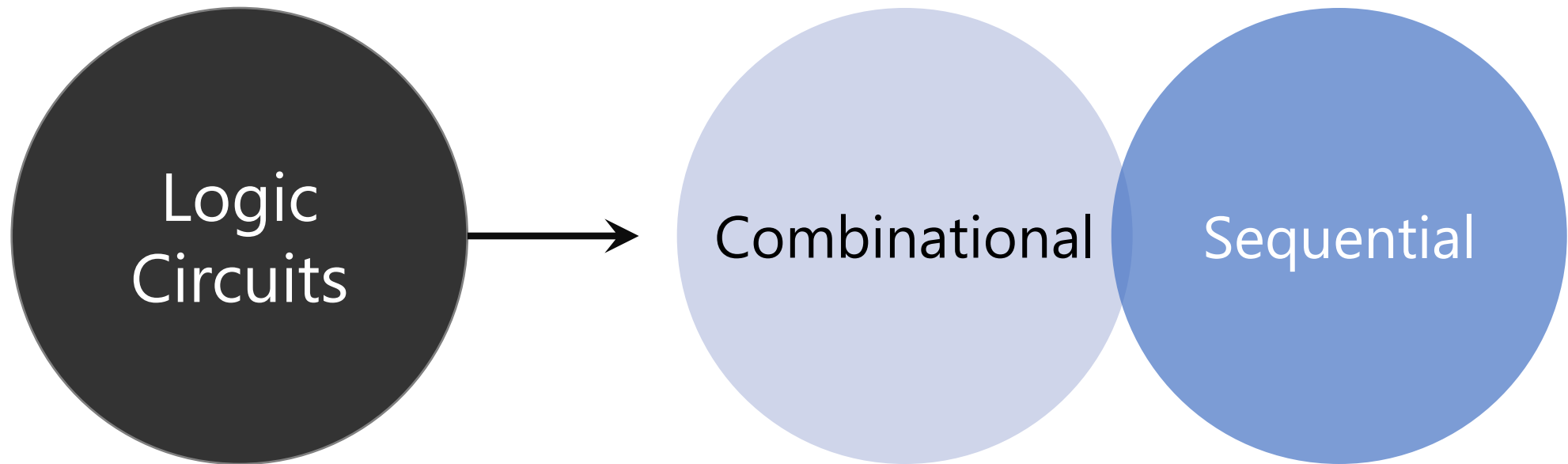


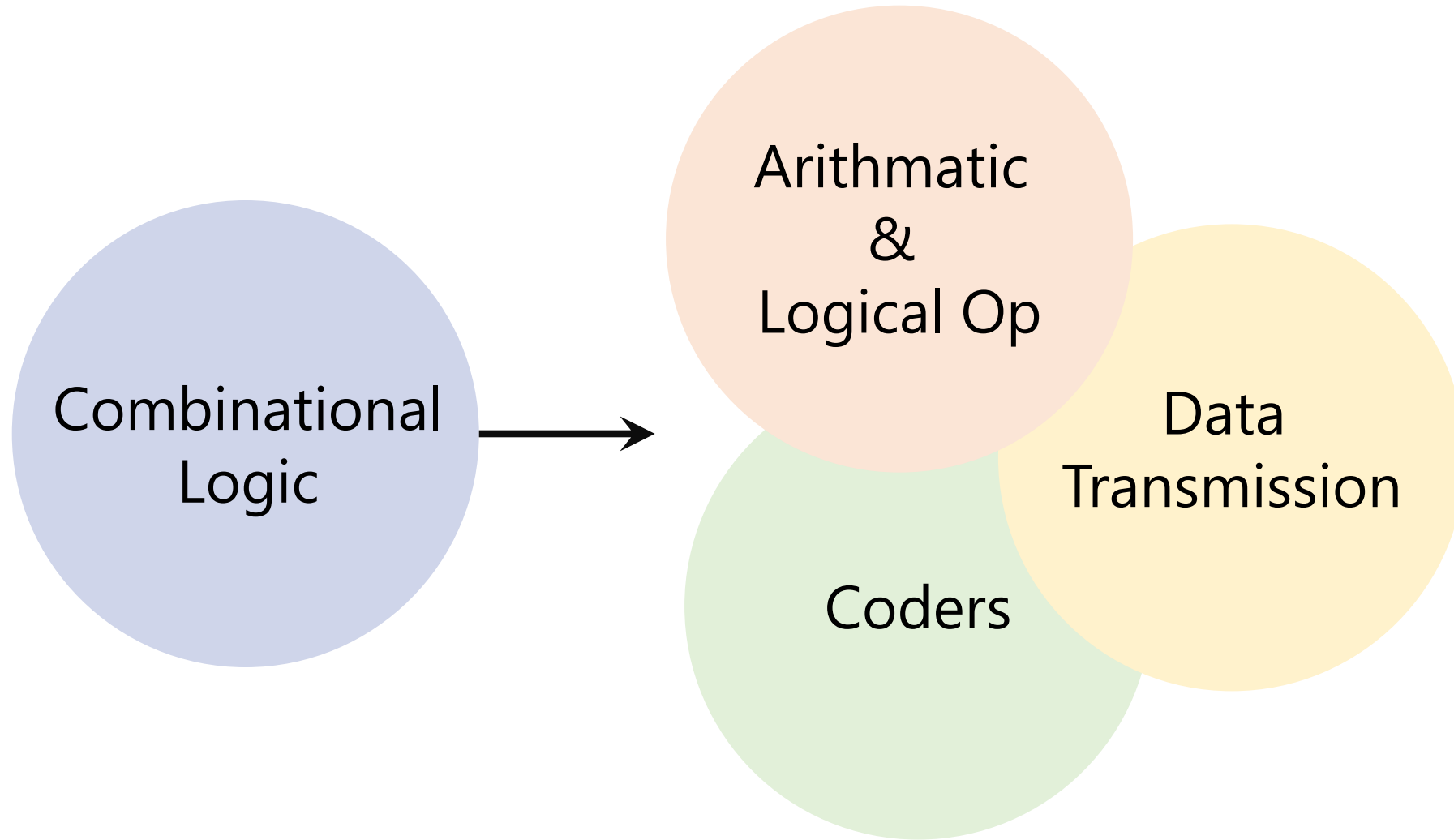
We have our ideal memory unit: Flip-Flop
Let's build sequential (logic) circuits!

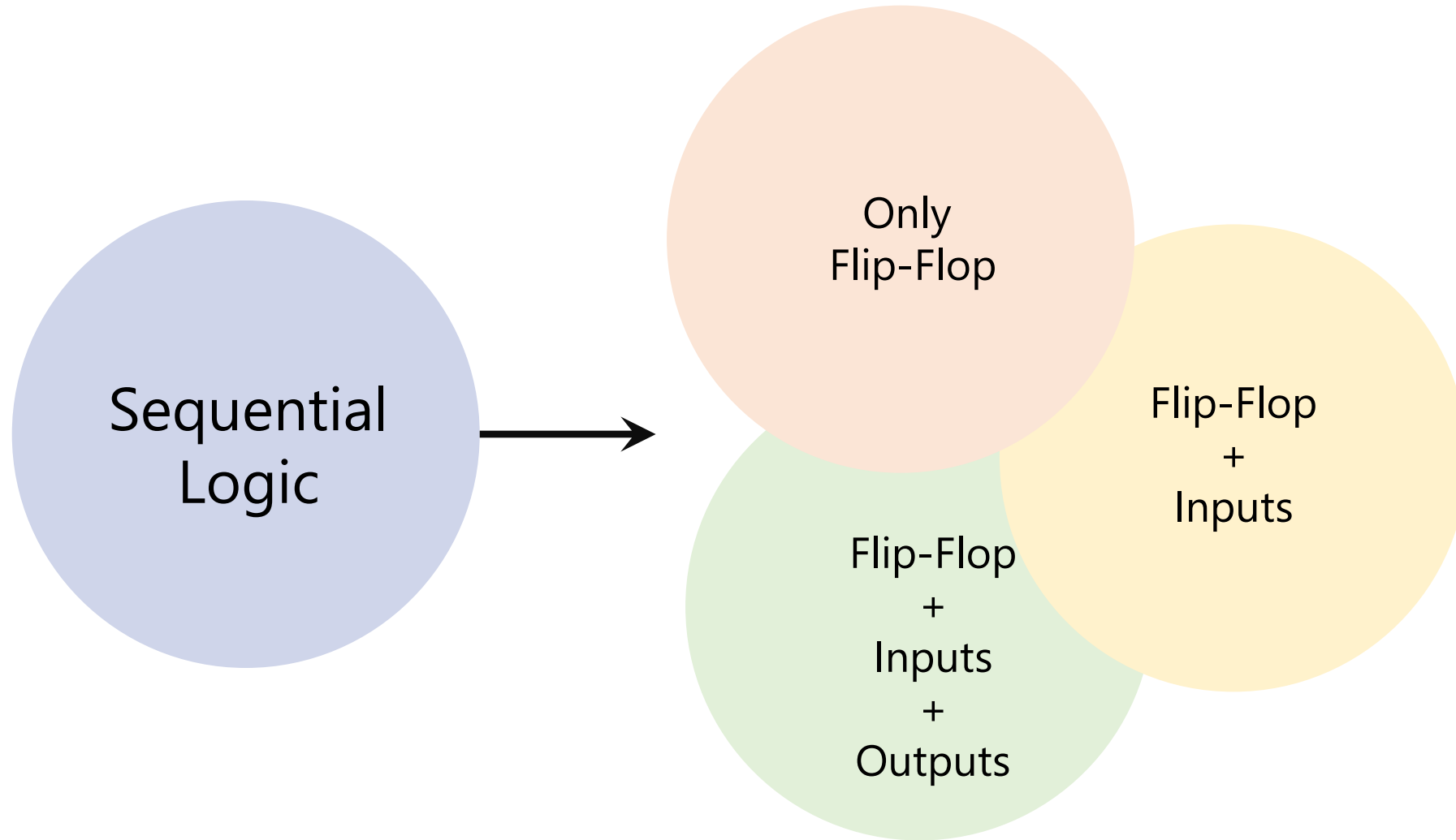
Analysis
vs.
Design

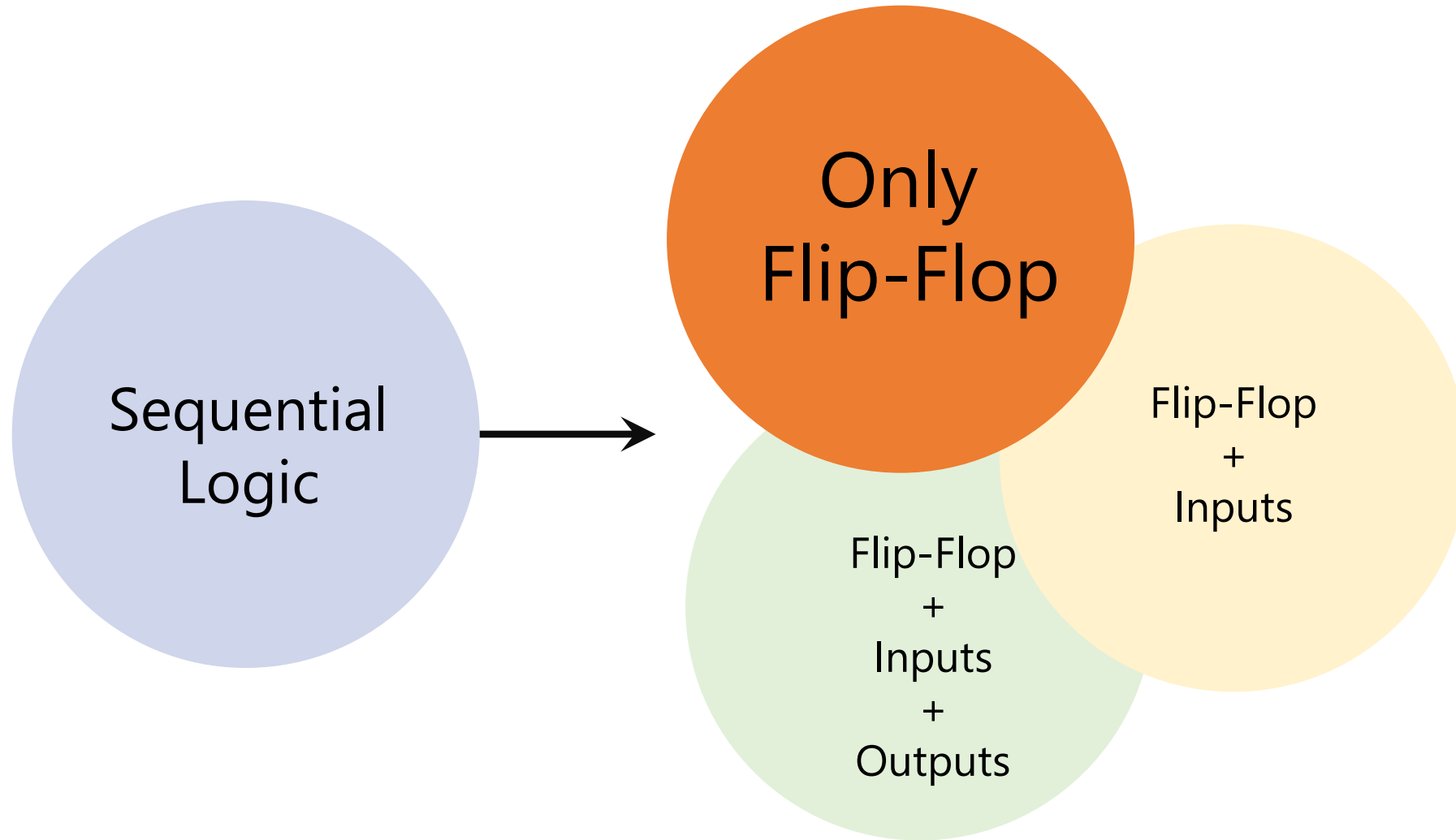
Analysis: Given a sequential circuit, show the behavior
vs.

Design: Given a behavior, build the sequential circuit

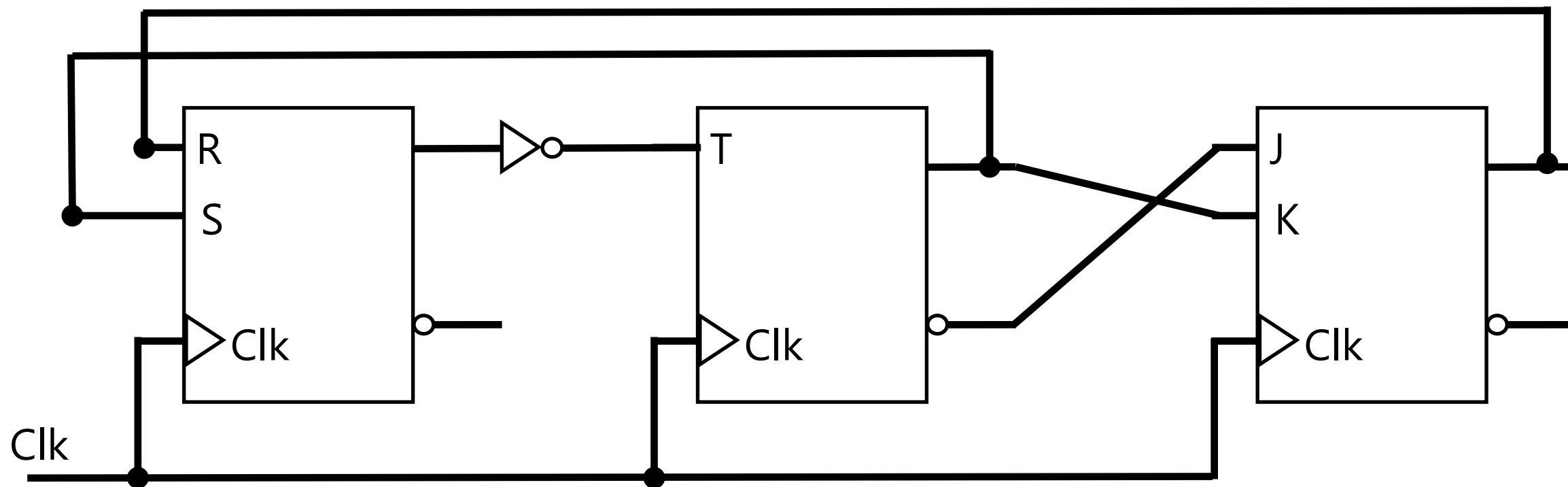








Analysis by an example



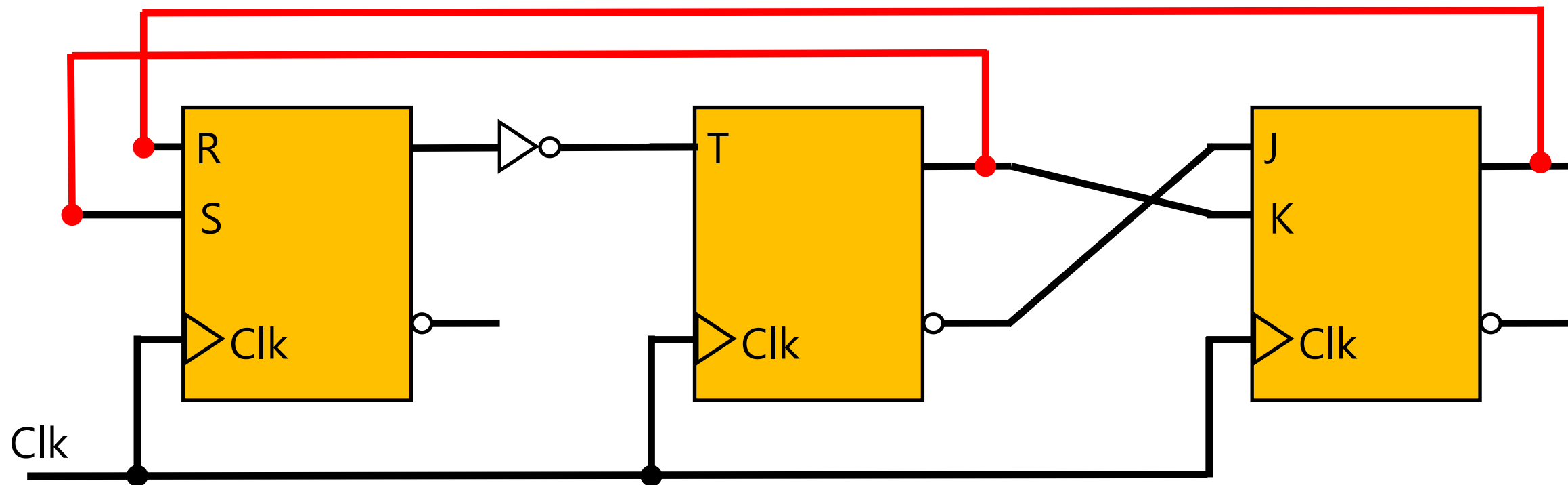
Analysis

0) Is it sequential circuit?

At least one FF $\rightarrow$ Yes

At least one feedback $\rightarrow$ Yes

Otherwise $\rightarrow$ No

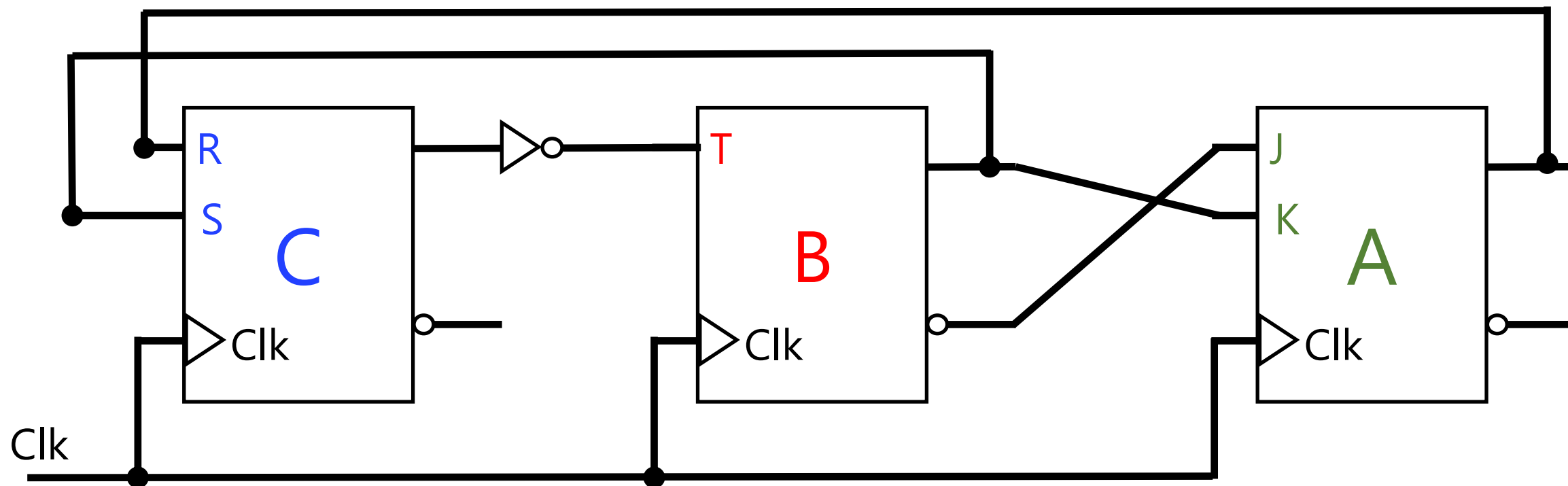


Analysis

1) What are the FFs?

1.1. We pick a name for each FF

1.2. We note the type of FF

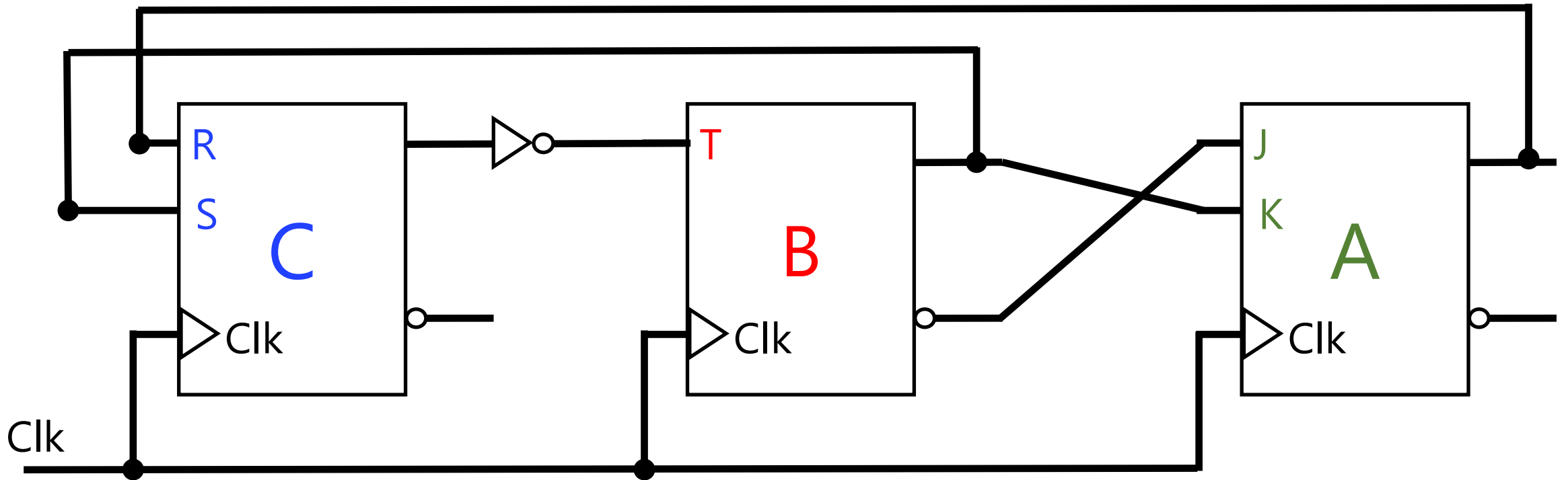


Analysis

2) What are the state combinations
(possibilities)?

Each FF can have $\{0,1\}$ states

In total, $2^{\text{\#FFs}}$



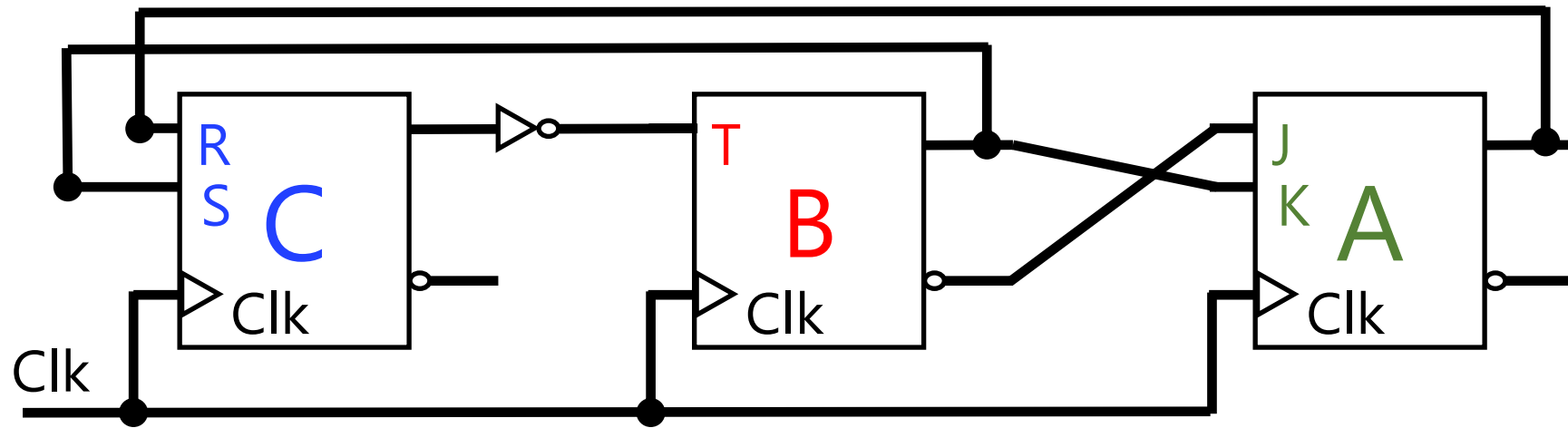
#FFs = 3 $\rightarrow 2^3 = 8$ combinations

Analysis

3) Form a 'State' Table

3.1. For each FF, one column for **current** state

3.2. For each FF, one column for **next** state

[illegible]

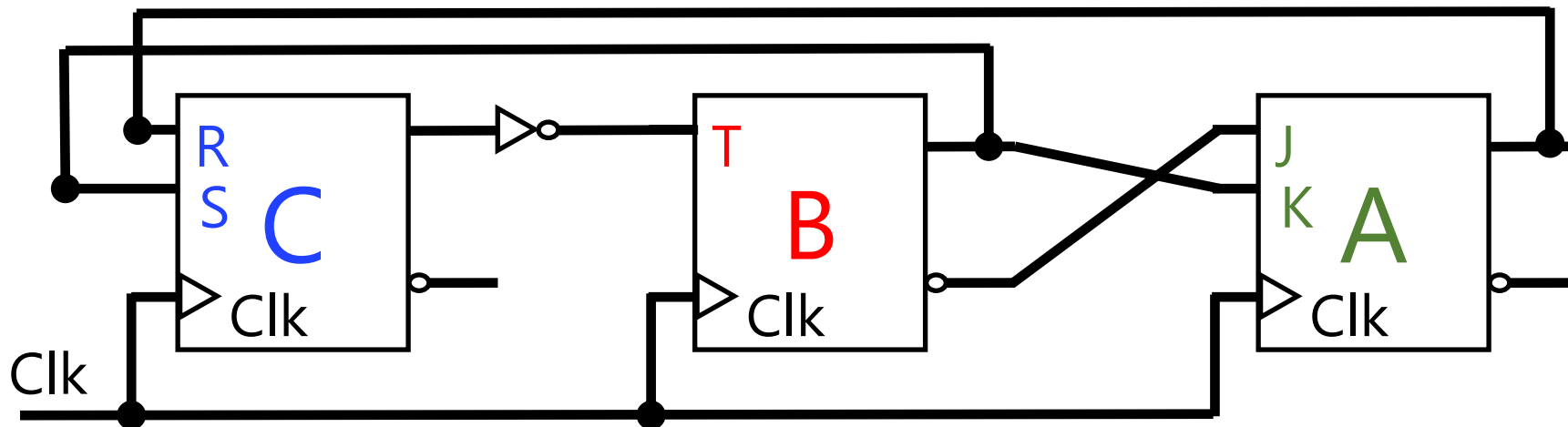
Analysis

3) Fill the 'State' table

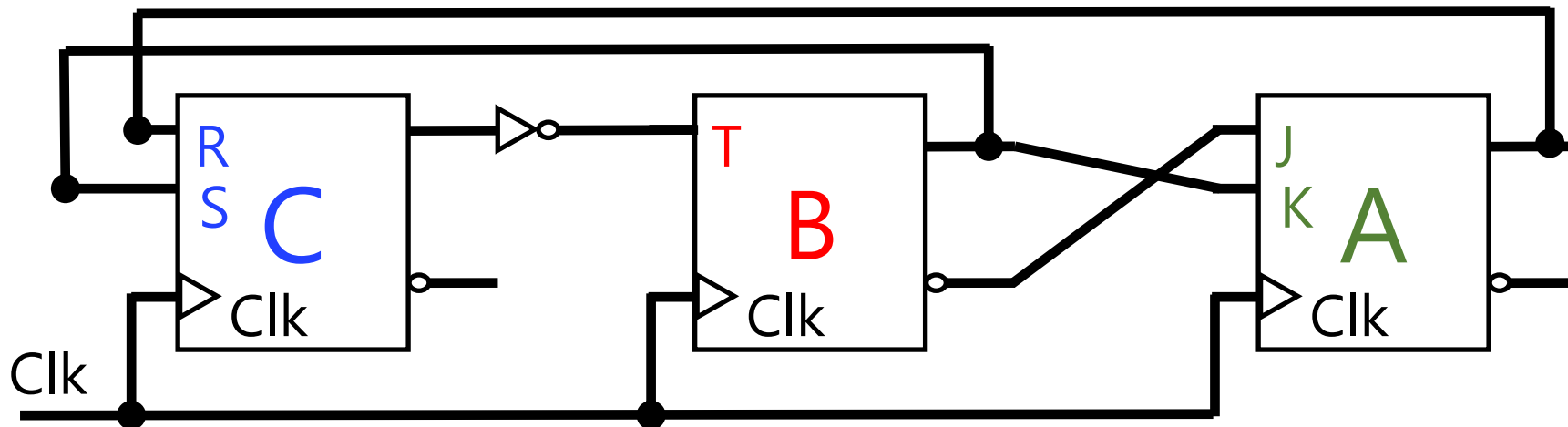
For each FF, we determine the **next** state based on

I) current state

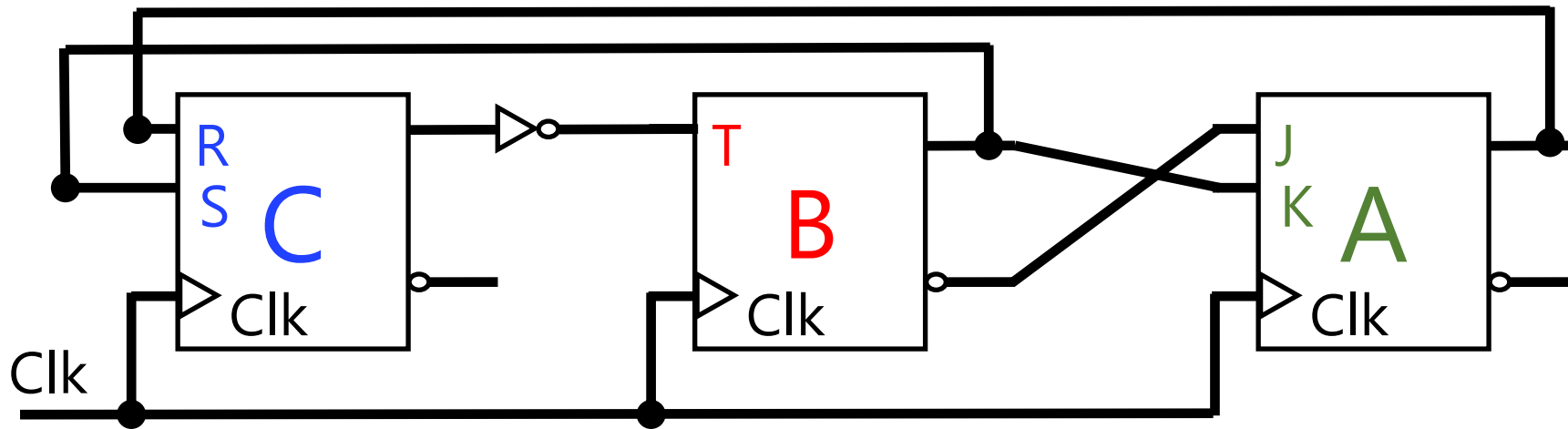
II) the current value of inputs to the FF



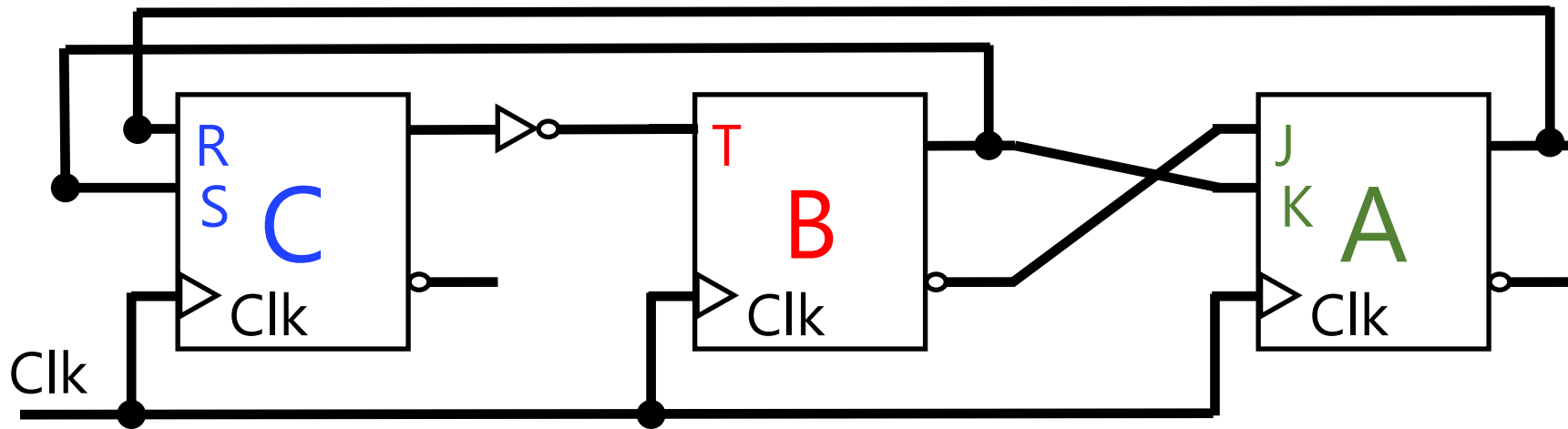
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



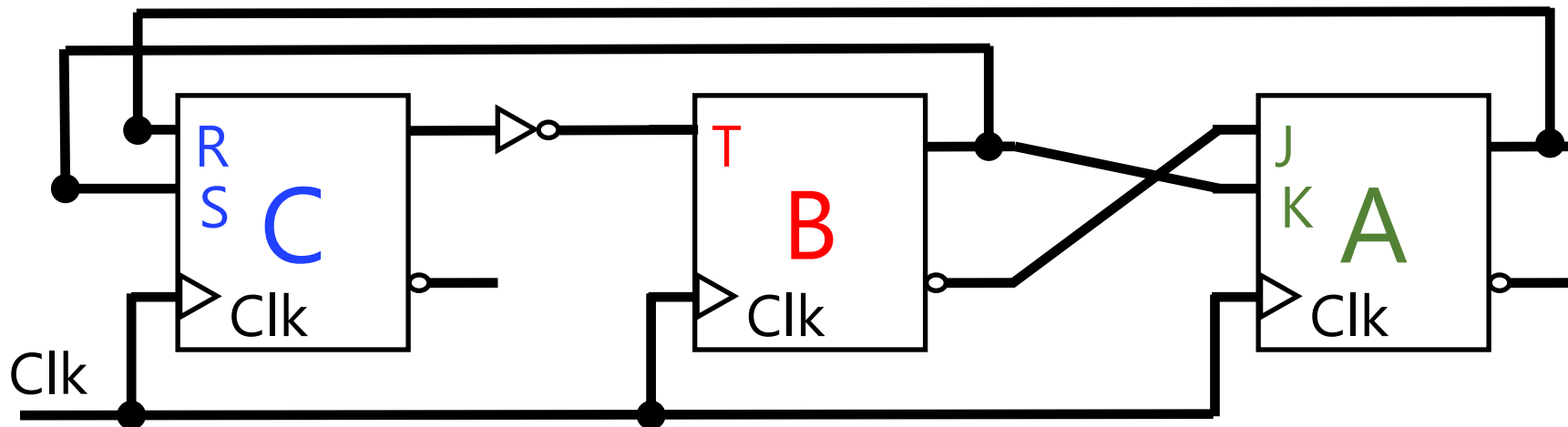
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			?
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



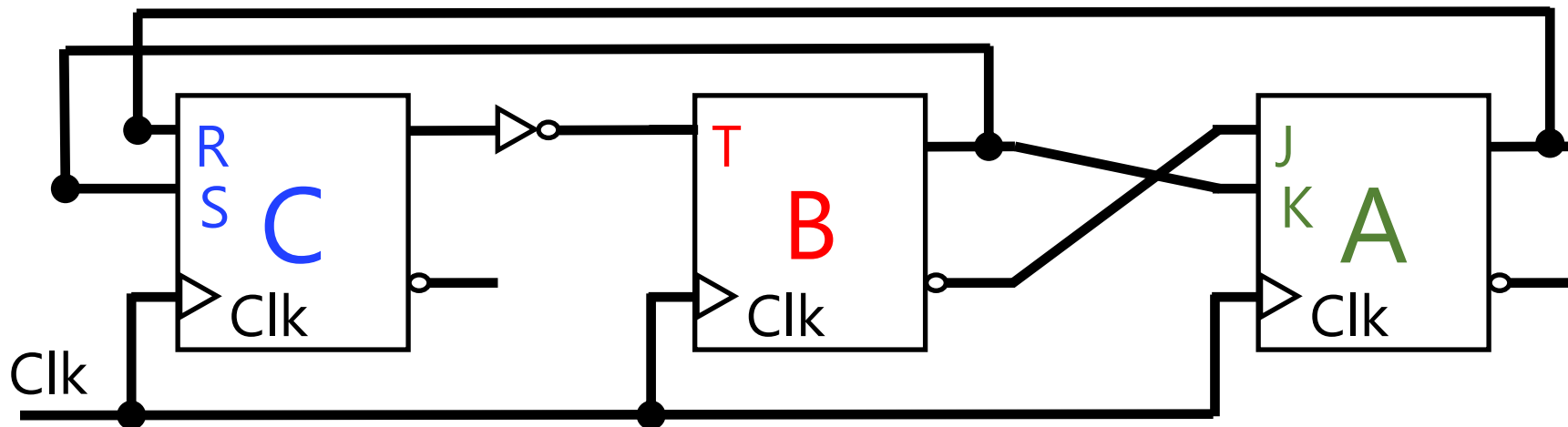
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			$Q_A(T)=0$ $J_A=Q'_B(T)=1$ $K_A=Q_B(T)=0$
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			



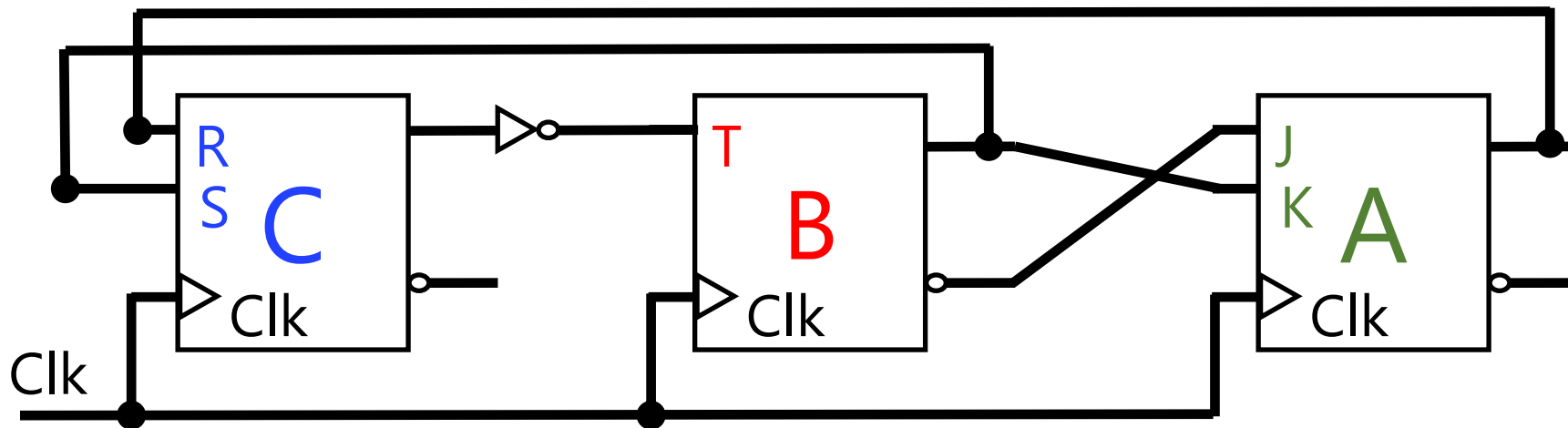
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			$Q_A(T)=0$ $J_A=Q'_B(T)=1$ $K_A=Q_B(T)=0$ ----- Set Action: 1
0	0	1			
0	1	0			
0	1	1			
1	0	0			



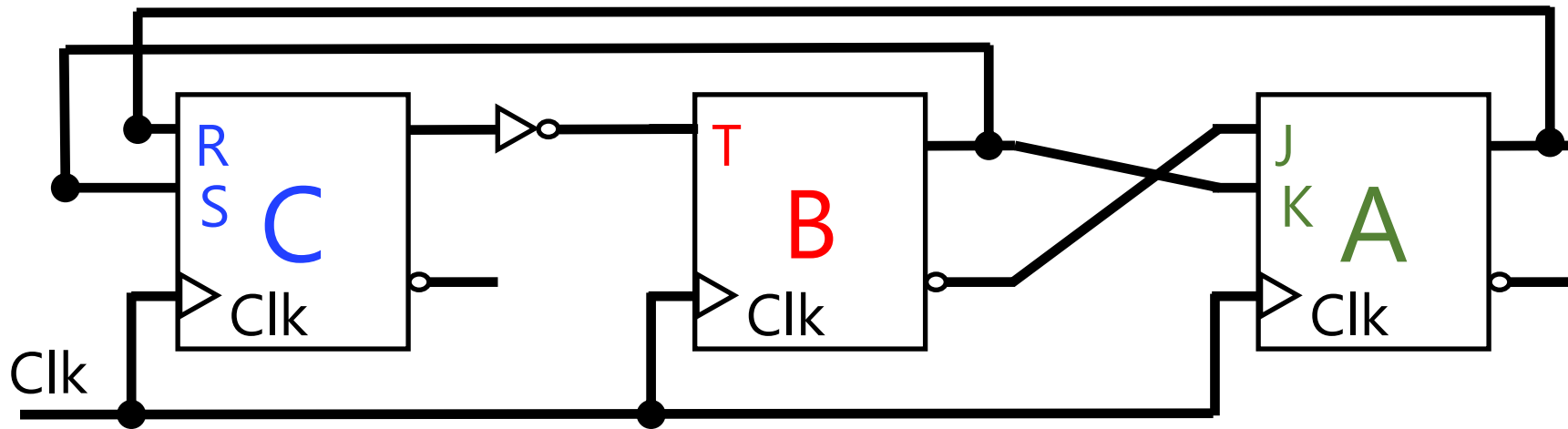
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



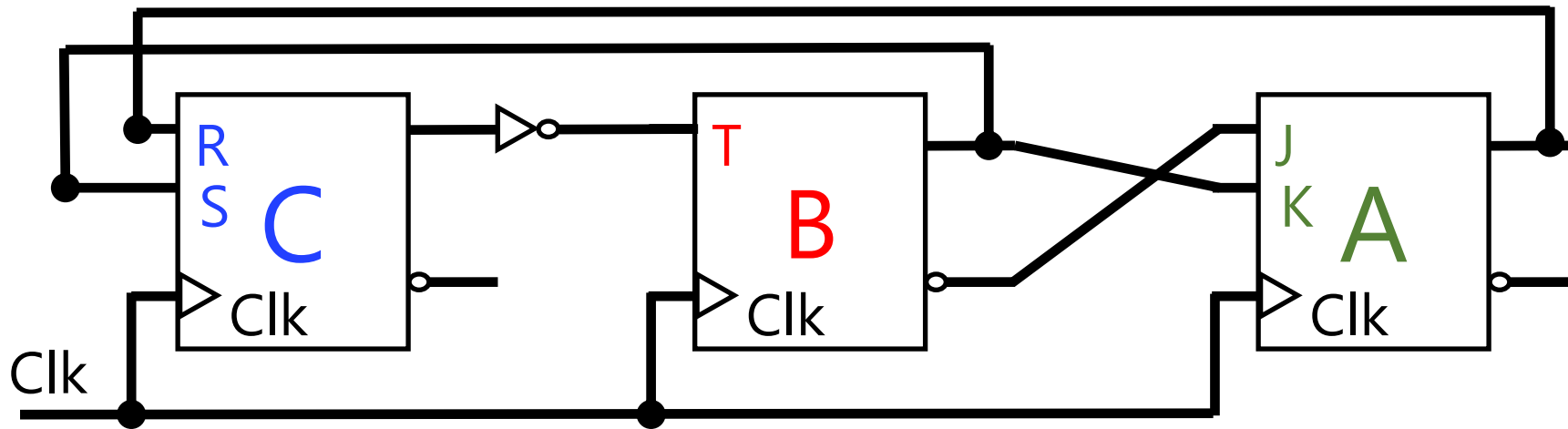
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0		?	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



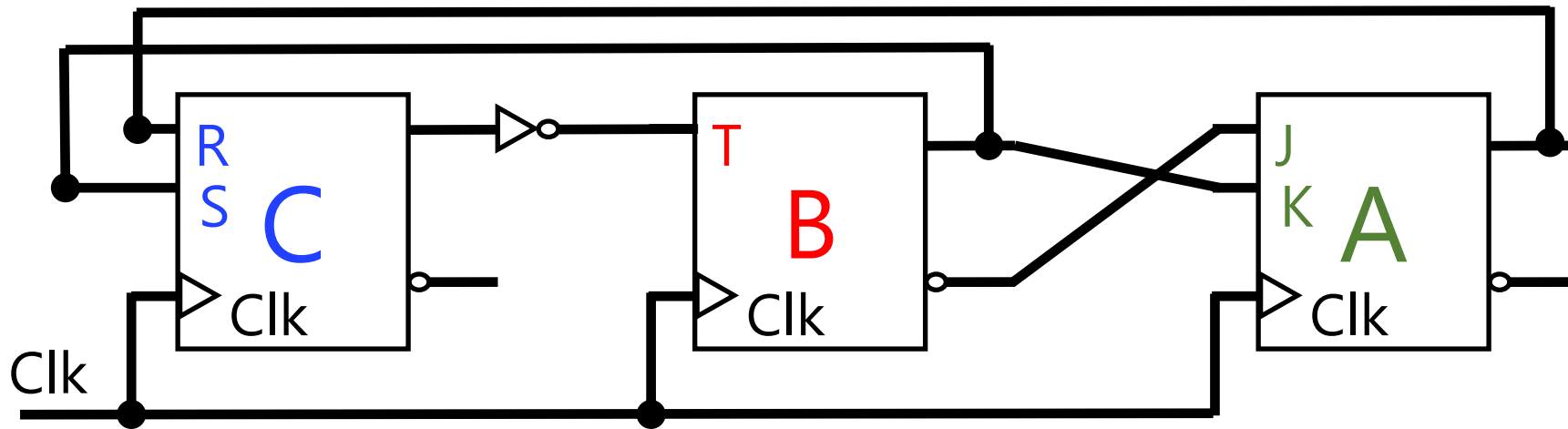
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0		$Q_B(T)=0$ $T_B=Q'_C(T)=1$	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			



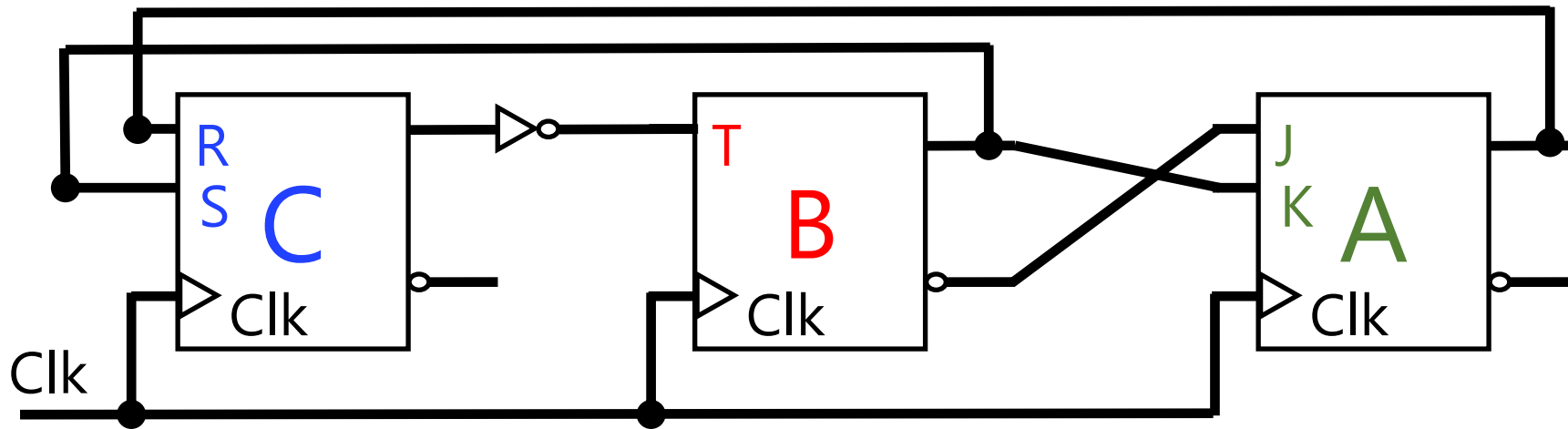
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0		$Q_B(T)=0$ $T_B=Q'_C(T)=1$ ----- $\text{Comp. } (Q_B(T)) = 1$	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			



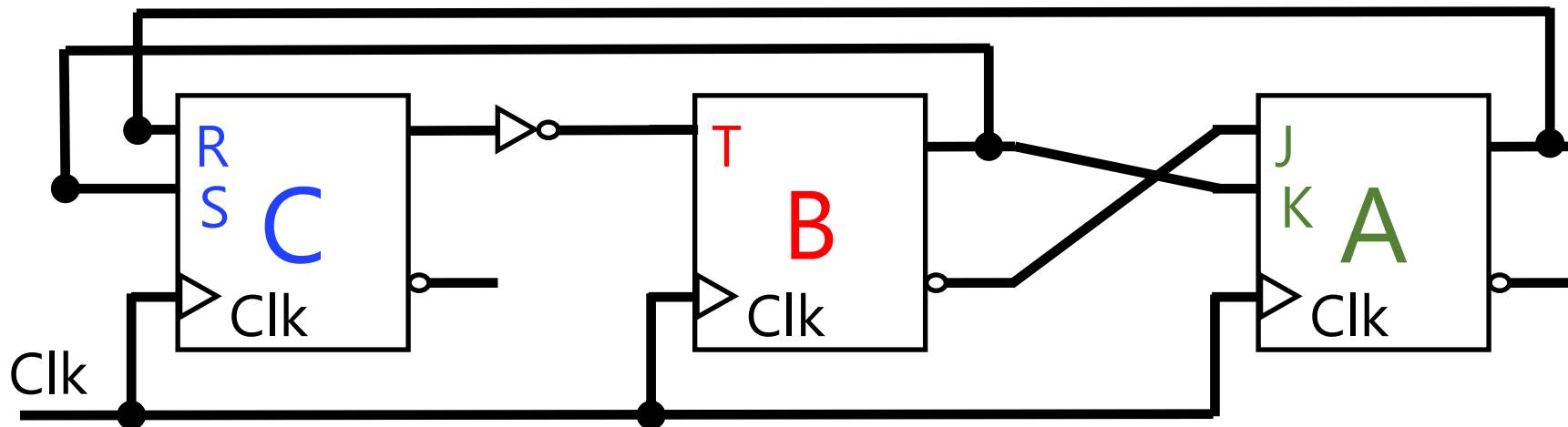
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	?	1	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	$Q_C(T)=0$ $R_C=Q_A(T)=0$ $S_C=Q_B(T)=0$	1	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	$Q_C(T)=0$ $R_C=Q_A(T)=0$ $S_C=Q_B(T)=0$ ----- Store $Q_C(T) = 0$	1	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			

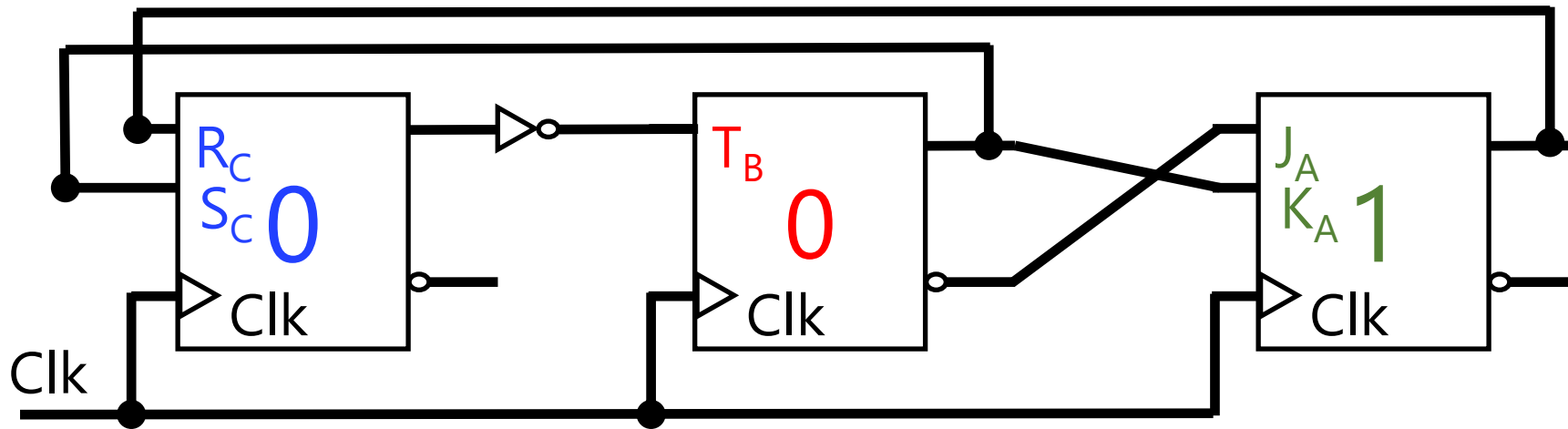


Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

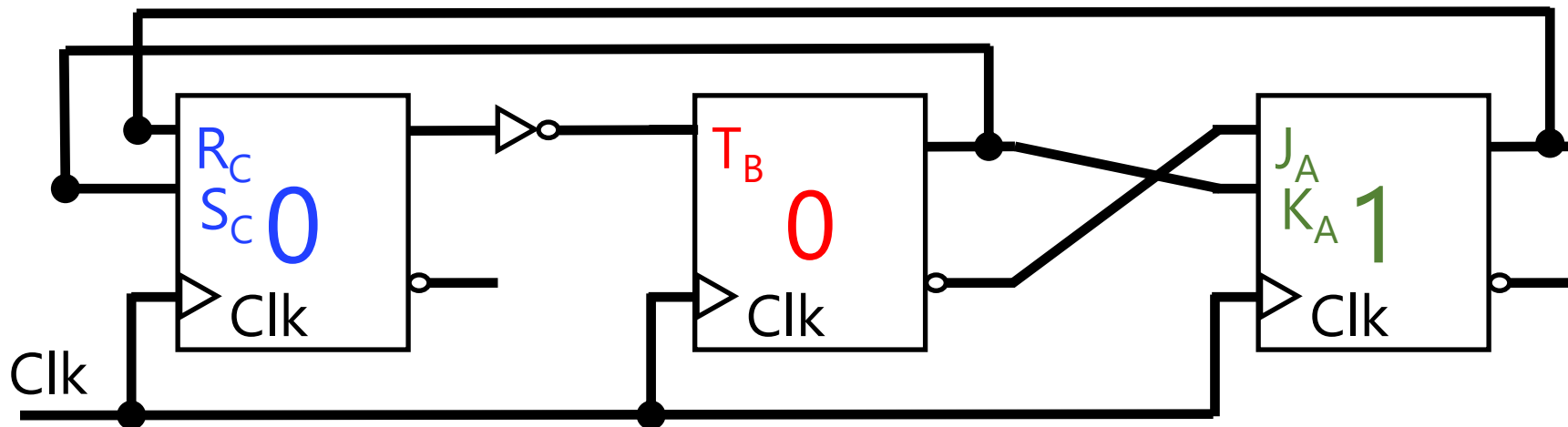
Analysis

$$Q_A(T) = A, Q'_A(T) = A'$$

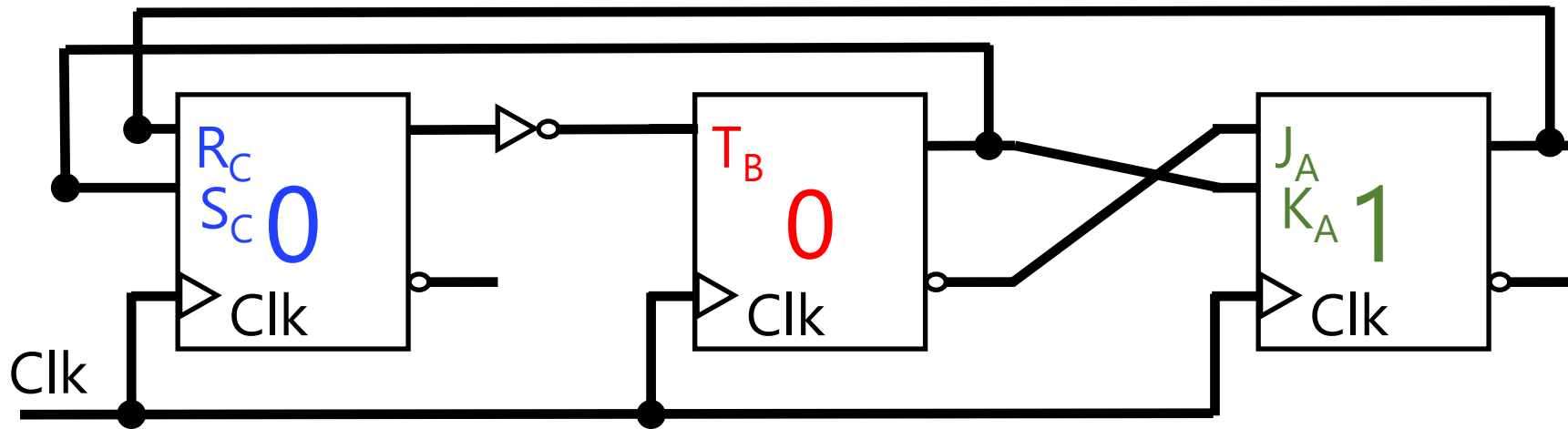
For simplicity, the **current status** of a FF can be assume to be as a **binary variable**



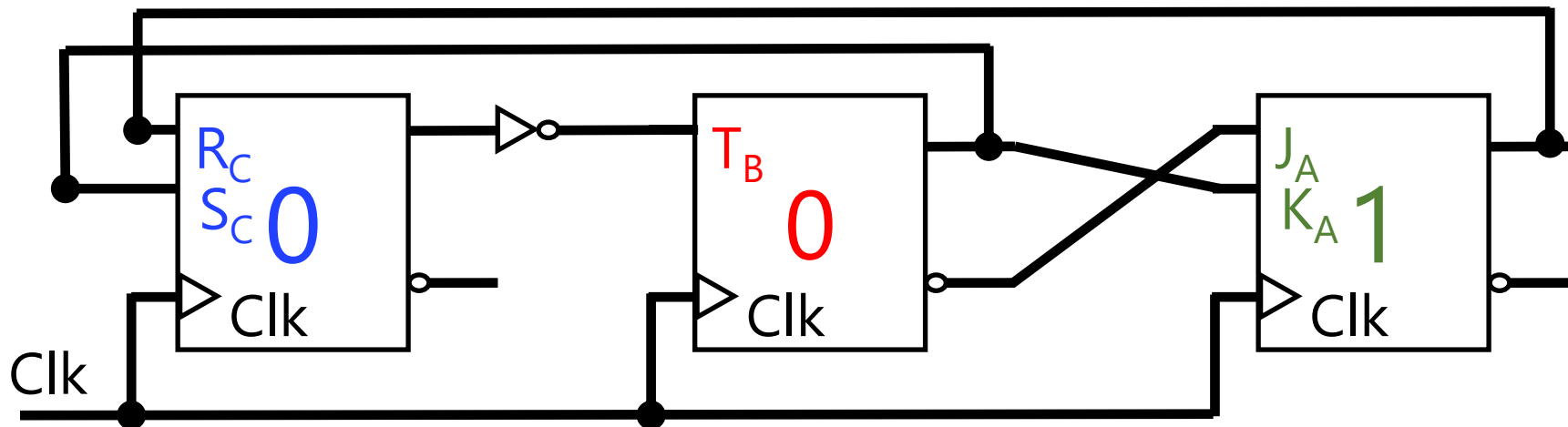
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1			A=1 $J_A = B' = 1$ $K_A = B = 0$ ----- Set Action: 1
0	1	0			
0	1	1			
1	0	0			



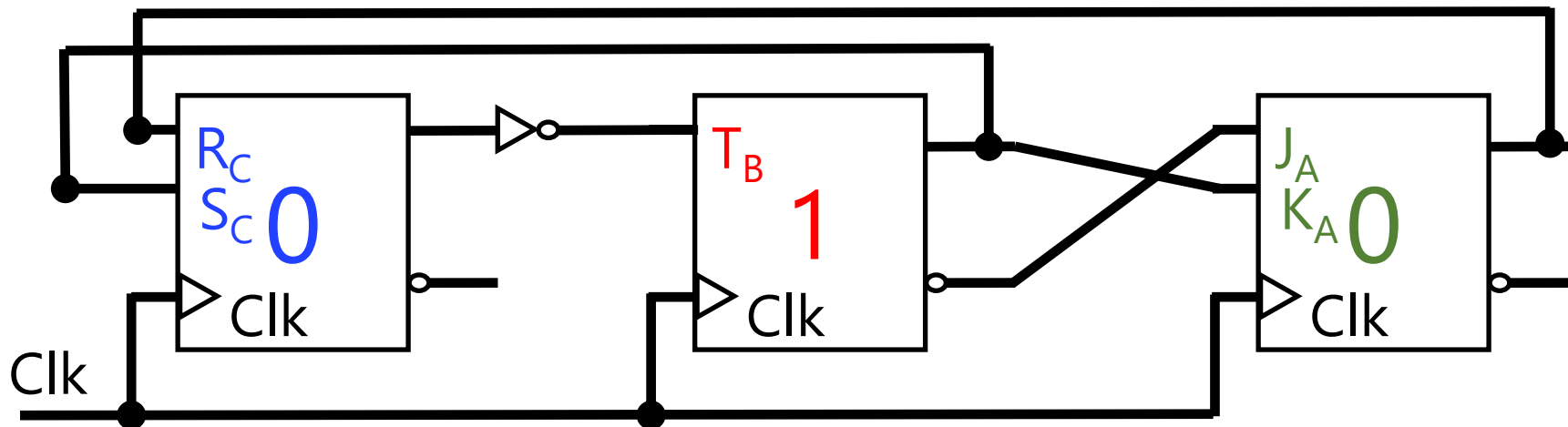
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1		B=0 T _B =C'=1 ----- Comp. Action: 1	1
0	1	0			
0	1	1			
1	0	0			
1	0	1			



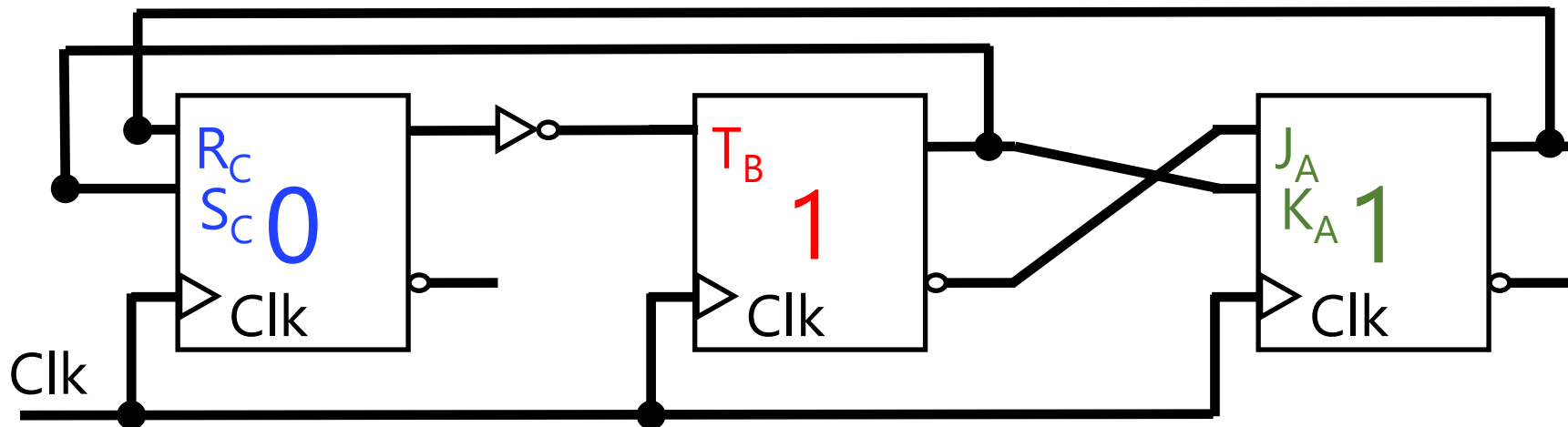
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	C=0 R _C =A=1 S _C =B=0 ----- Reset Action: 0		
0	1	0			
0	1	1			
1	0	0			



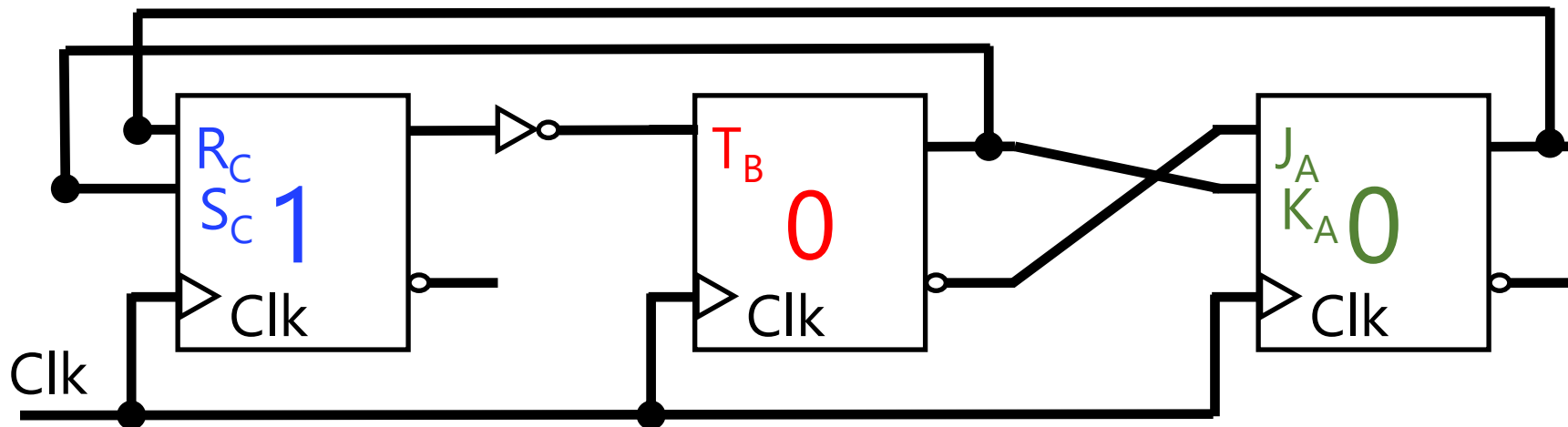
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



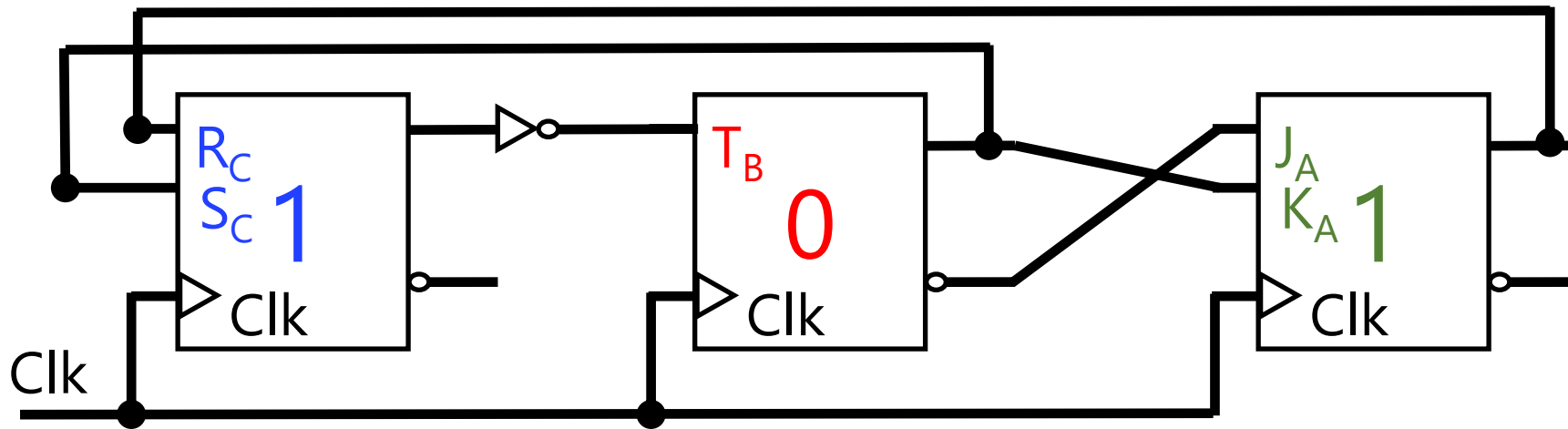
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



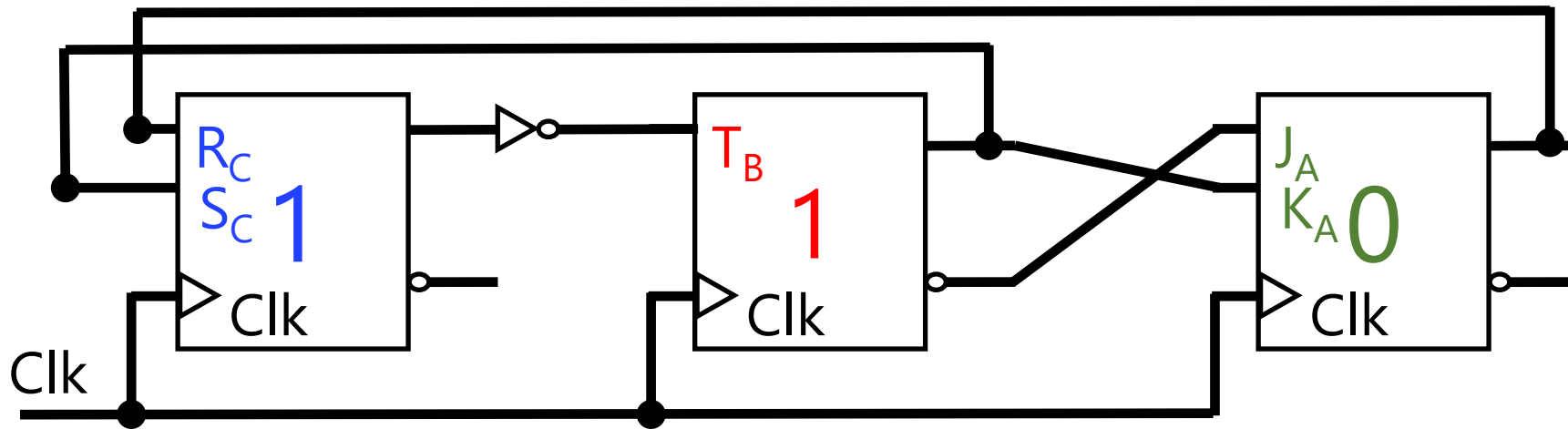
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	x	0	0
1	0	0			
1	0	1			
1	1	0			
1	1	1			



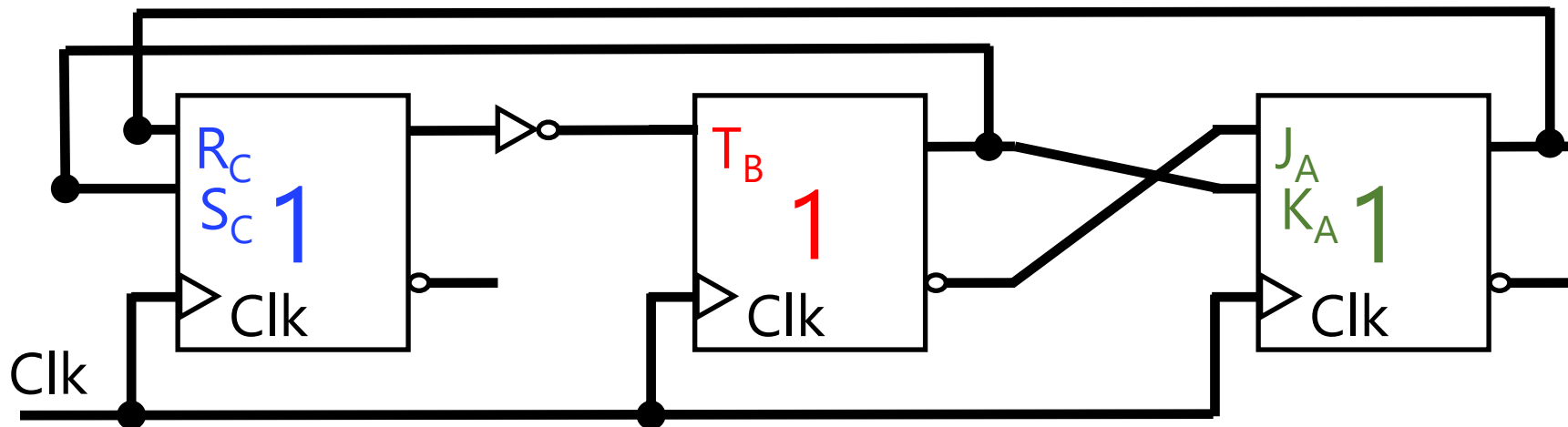
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	×	0	0
1	0	0	1	0	1
1	0	1			
1	1	0			
1	1	1			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	x	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0			
1	1	1			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	×	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	×	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	×	1	0

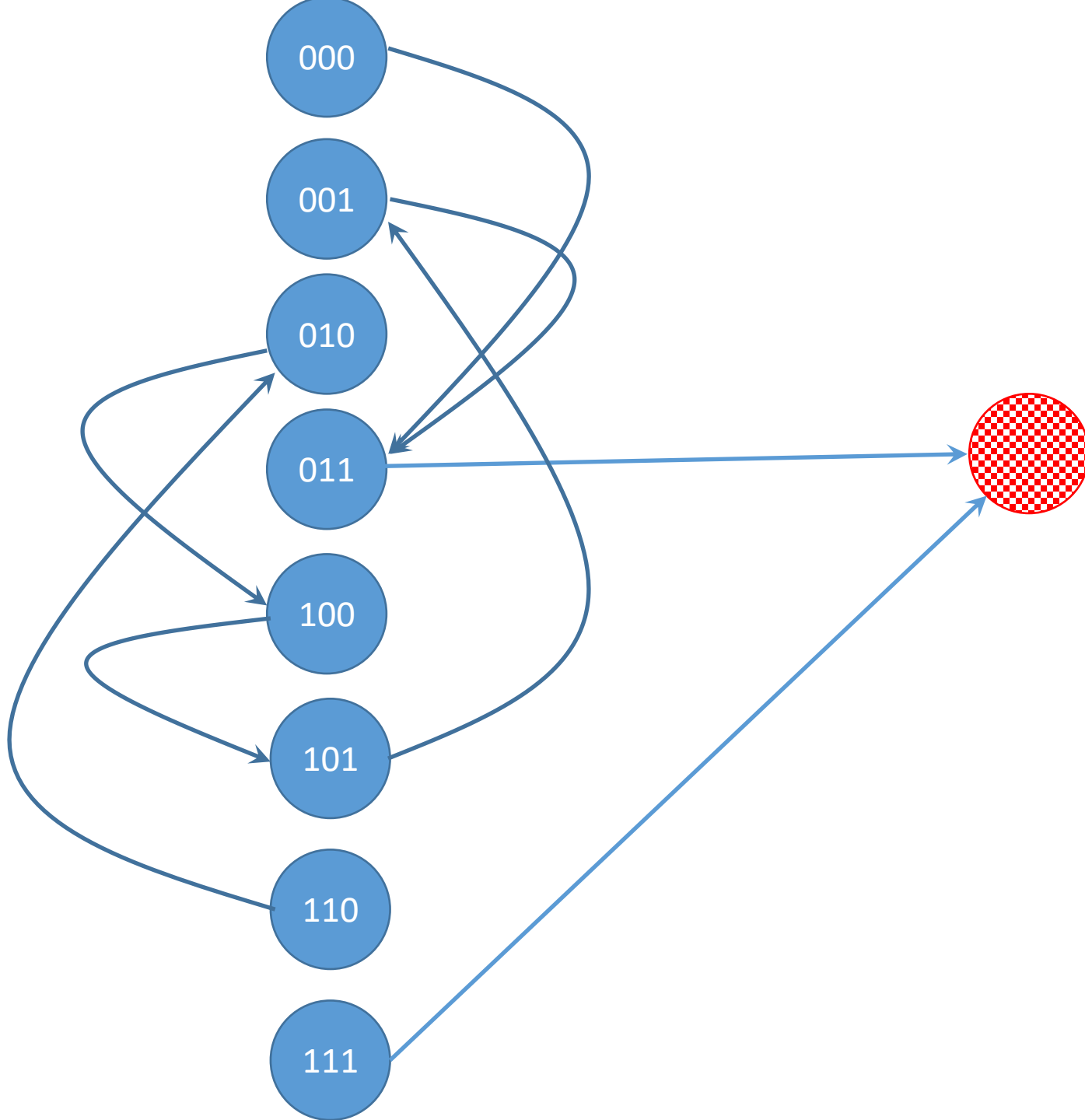
Analysis

4) State Transition Diagram

4.1. for each state combination (each row), a node

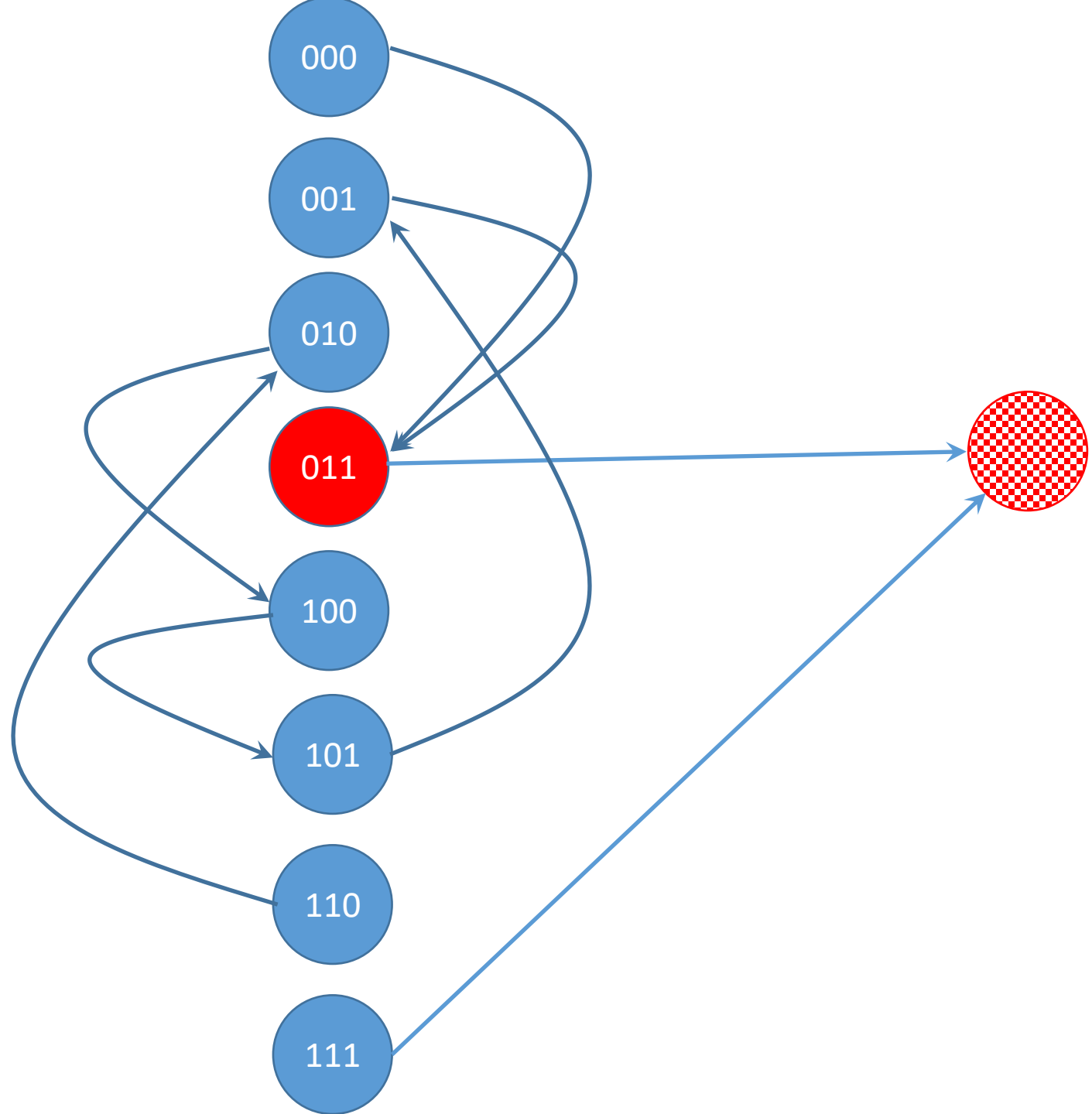
4.2. from one state (node) to another state, a directed edge

CBA



Analysis

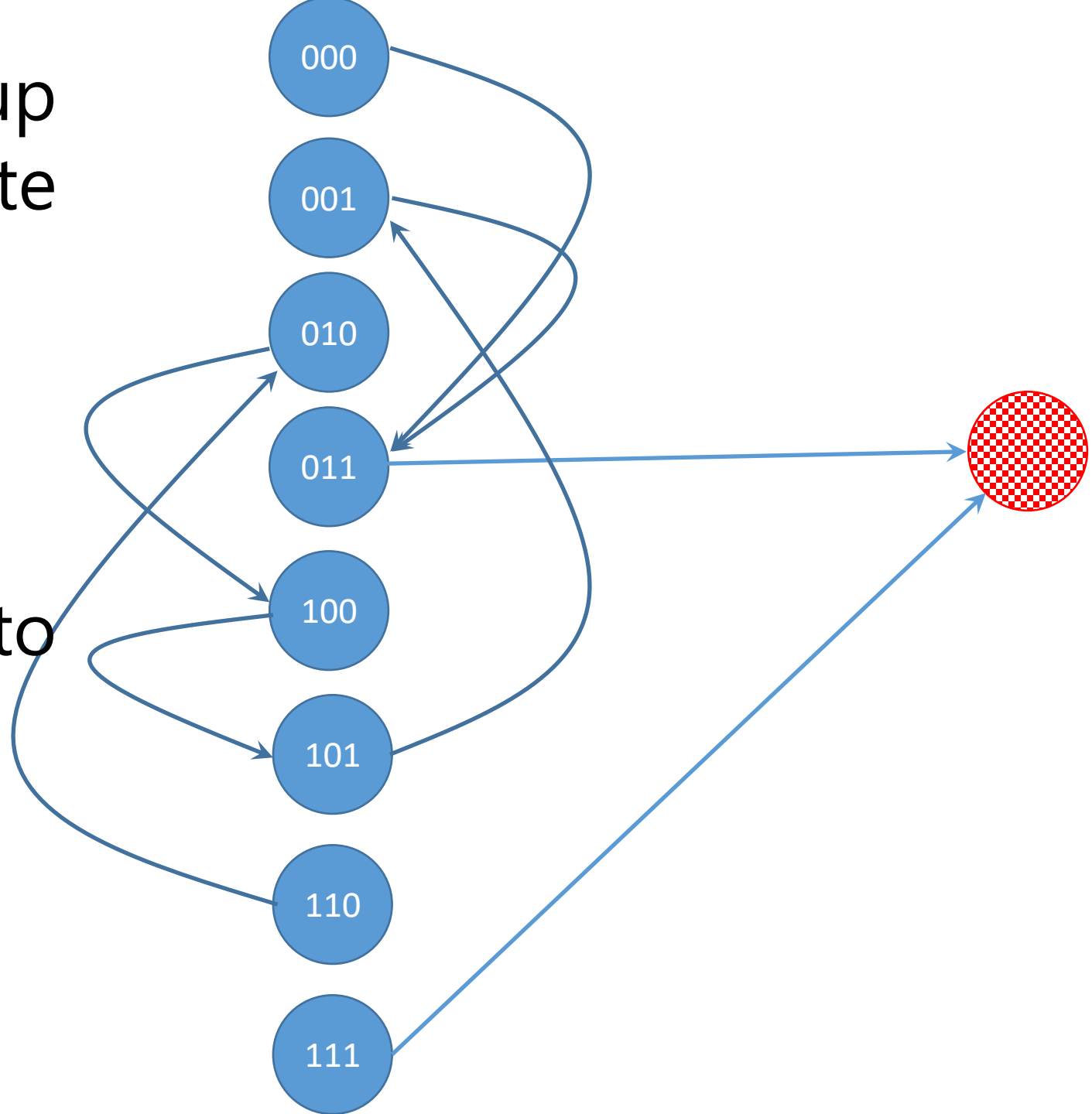
5) (*Optional*) Path on State Transitions

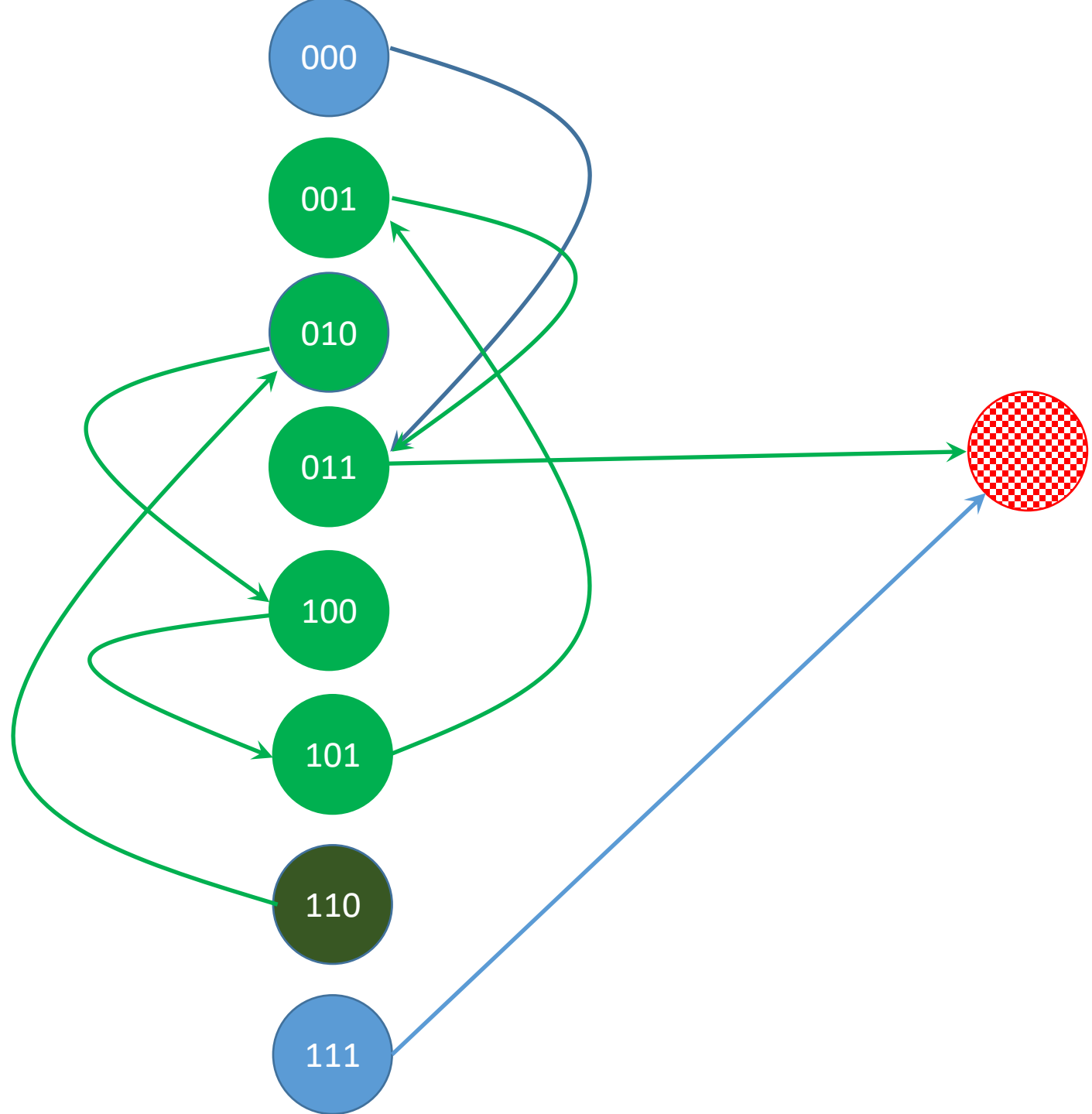


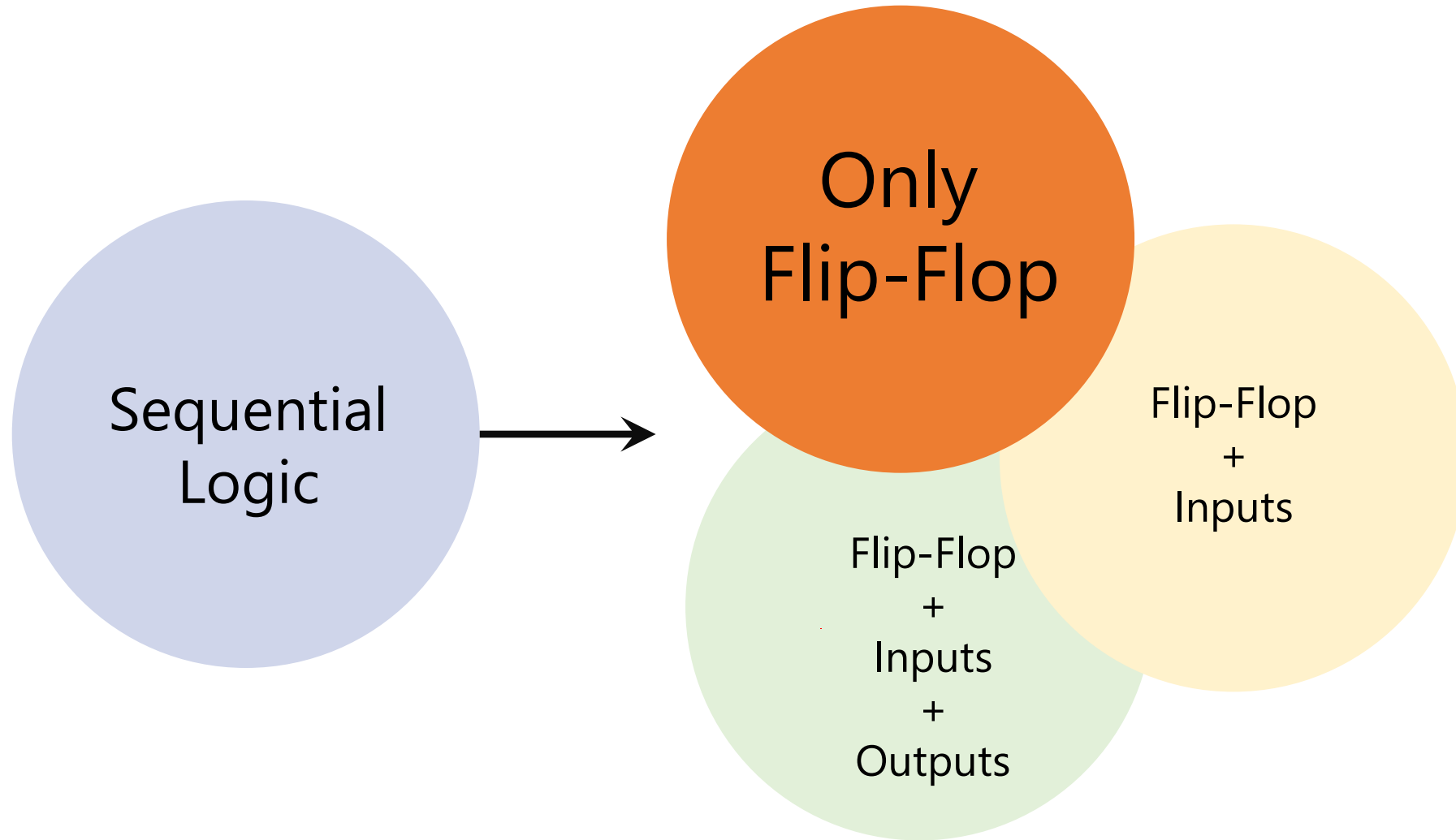
All the paths end up
with indeterminate
state



The circuit needs to
be improved!







Analysis (Recap)

0. Is the circuit sequential or combinational? Any FF or feedback → Sequential
 1. What are the flip-flops? RS, D, T, JK, or mixed (e.g., 2 JK, 1 RS, ...)
 2. What are the state combinations? $2^{\#FF}$
 3. Form "State" table:
 - a) Columns: for each FF, two columns:
 - one for current state,
 - one for next state
 - b) Rows: for each state combination
 - In total: $2^{\#FF}$
 4. Fill the state table for next state columns based on:
 - a) the current state
 - b) the inputs to the FFs
 5. Form State Transition Diagram
 6. (Optional) Analyze paths and states in state transition diagram
-

Design by an example
